



Physically driven feature engineering for deep learning applications in seismo-volcanic signal analysis

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Abstract

The progressive growth of seismological databases has motivated the exploration of novel methodologies for common tasks such as detection and phase-picking, with a focus on maintaining reliability comparable to human performance. This goal consistently involves leveraging deep learning techniques, which emulate sensory processing in the human brain through numerical simulations. This study introduces a physically driven feature engineering approach that capitalizes on the inherent information within seismic data. While many contemporary studies train their models via robust raw datasets, practical alternatives tailored for smaller databases are often overlooked. Feature engineering in seismological contexts aims to develop deep learning models with tangible physical significance, specifically those that target event detection and phase-picking tasks across both local and regional seismic environments. Our approach leverages physically driven feature transformations for the joint detection and phase-picking task. This includes incorporating the energy signal envelope for effective seismic event classification, using amplitude spectra from signals filtered at predefined frequency bands, and calculating spatial features (such as wave incidence and azimuth) for accurate phase-picking. This integrated feature set optimizes model performance, especially when dealing with small volcanic seismology datasets. The proposed joint methodology is particularly pertinent in seismo-volcanic contexts, where accurate discrimination and characterization of seismic signals are pivotal for monitoring and risk assessment purposes. The incorporation of significant physical information from seismic signals into pattern recognition is crucial, as many feature engineering applications lack a contextual understanding of the data, which can lead to distortions, particularly within geophysical domains. Our results demonstrate human-level performance in these common tasks, harnessing the capabilities of statistical learning algorithms as a practical, resource-efficient solution for addressing these challenges on a large scale.

Keywords Deep learning · Feature engineering · Seismic detection · Seismic phase-picking

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Introduction

During the last two decades, the density of global and regional seismic networks has increased; three-component seismic signals are now available for conventional seismic stations; and more recently, the distributed acoustic sensing (DAS) technique has contributed to the acquisition of larger datasets. Seismology is evolving into a big data problem, and traditional tasks in seismic or volcanic observatories, such as seismic event detection, wave phase picking, localization, and focal mechanism determination, which are generally highly dependent on the analysts, will benefit from deep learning advances.

Seismology has undergone important changes in terms of volume, variety, and velocity in the last few years. In addition, machine and deep learning applications integrate a great diversity of data to exploit the potential of finding complex patterns and significant characteristics, which are computationally efficient and accurate at the human-level (Kong et al. 2019). Deep learning models have found widespread application in data-driven activities, including seismology. Recent works on seismology have employed convolutional neural networks (CNNs) (Perol et al. 2018; Ross et al. 2018), recurrent neural networks (RNNs) (Shan et al. 2020), hybrid CNN and RNN models (Mousavi et al. 2019a, b; Ross et al. 2019; Soto and Schurr 2021; Zhou et al. 2019), encoder-decoder architectures (Zhu and Beroza 2019), and transformers (Feng et al. 2022; Mousavi et al. 2020). Many of these applications involve direct engagement with raw data and undergo minimal preprocessing before model training and testing.

Using raw data to train robust architectures for pattern extraction is a key strength of deep learning approaches. For example, seismic catalogs such as STEAD (Mousavi et al. 2019a, b), LenDB (Magrini et al. 2020), and GEO-FON datasets (Hanka and Kind 1994), comprising millions of waveforms, provide samples of P-waves, S-waves, and noise. These datasets cover a wide range of earthquake focal mechanisms and geological settings, facilitating the development of standardized procedures for statistical learning tools and ensuring homogeneous access to data. Notable examples include the SeisBench seismological benchmark (Woollam et al. 2022).

However, relying solely on raw data can be computationally expensive and may result in diminished performance when working with poor-quality or small datasets. Additionally, deep learning models often face inherent uncertainties, as the search for optimal filters during the training phase lacks explicit guidance. For data tied to the physical world, identifying a set of numerical features that provide a more realistic representation of real-world phenomena becomes

increasingly important. Therefore, exploring alternative strategies to increase model training effectiveness is crucial.

In the context of statistical learning, two primary approaches address the challenges posed by limited or low-quality datasets: data augmentation and feature engineering. These methods provide viable pathways to improve model performance and reliability under constrained conditions. Feature engineering involves a range of processing algorithms designed to extract meaningful attributes from raw data, with a focus on identifying the most relevant aspects of inputs prior to the training phase. This process aims to improve the accuracy of the output results. These attributes, commonly referred to as *features*, capture a wide variety of characteristics of the observed phenomena, which may be statistical or physical in nature (Dong and Liu 2018). Through feature engineering, analysts determine optimal strategies for data manipulation, such as reducing the dimensionality of the feature space, representing diverse input types, and applying suitable normalization for specific datasets. This technique has a significant effect on statistical learning models and their predictions. While feature engineering can be applied to both small and large datasets, its benefits are especially pronounced when working with small datasets or low-quality data. The overarching goal is to extract the most relevant aspects of input data to improve the accuracy and performance of predictive models. The effectiveness of feature engineering procedures varies depending on the context of the model and the nature of the data (Zheng and Casari 2018).

Feature-based representations of time series data offer significant advantages by improving model optimization, reducing computational costs, and providing relevant advantages especially when training data is scarce. Instead of relying on increasingly complex model architectures in domains that may not adequately represent the data for a specific application, transforming time series data into feature space enhances model accuracy (Dong and Liu 2018). For geophysical data, features extracted from the frequency or time-frequency domains are particularly valuable. These features play a critical role in processing and analysis tasks, making them especially important for optimizing statistical learning through feature engineering.

Few studies have focused on temporal data feature extraction, but some applications exist in areas such as retail analysis, biomedical signal processing, and human motion measurement. For example, Li et al. (2018) conducted research on feature extraction from consumption records for retail behavior analysis. Their objective was to train recurrent neural networks on long short-term memory (LSTM) cells and support vector machine (SVM) models. This study emphasized multifeatured fusion and feature ordering to enhance model performance.

In the biomedical domain, Jaiswal and Banka (2017) applied feature extraction techniques on the basis of local pattern transformations. Their approach aimed to reduce computational resource demands in applications such as facial recognition, signal processing, and electroencephalogram (EEG) signal classification. Similarly, significant contributions are found in human activity recognition (HAR) data analysis. Li et al. (2020) trained a neural network using the power spectral density (PSD) of linear and triaxial accelerations from common human movements. Their methodology resulted in a substantial reduction in computational costs and improved prediction accuracy. Notably, they reduced the feature space from nine dimensions to just two, demonstrating the efficiency of their approach.

In seismological applications, the work of Wang and Teng (1997) is considered one of the pioneering efforts in utilizing seismic features to enhance artificial neural network models, specifically for S-phase identification and picking. Mousavi et al. (2016) quantified the most effective seismic attributes for distinguishing induced micro-earthquakes, using neural networks and logistic regression. Kong et al. (2016) developed a portable seismic network based on smartphone measurements. They improved these seismic readings by incorporating three representative statistical attributes: the interquartile range of the acceleration vector, the maximum zero-crossing rate, and a cumulative measure of signal amplitude. Brykov et al. (2020) proposed a backward-forward feature construction method to identify the optimal combination of features for an earthquake prediction algorithm. They claimed that their approach retrieves the best subset from a dataset without requiring extensive computational resources. Similarly, Khan et al. (2020) addressed earthquake detection via real-time accelerometer signals from electronic devices such as smartphones. Their study evaluated various features in both static and dynamic contexts to identify the most suitable machine learning model.

Volcanic seismology studies are frequently constrained by the scarcity of large, labeled datasets, posing a significant challenge for data-intensive deep learning models. Previous works (Legrand et al. 2023; Pertou et al. 2024) delineate the inherent difficulty of accurately discriminating local seismic events from regional phases. To overcome both this data limitation and the event discrimination problem, this paper introduces a comprehensive methodology for seismic event detection and phase-picking that relies on a physically driven feature engineering approach. This strategy is crucial as it leverages established seismological knowledge to transform raw waveforms into robust, domain-specific features. By providing the model with highly informative input, this approach makes the chosen deep learning architecture highly appropriate and efficient despite the reduced

sample size inherent to our volcanic datasets. We focus on extracting significant seismic features, such as signal energy envelope, amplitudes for specific frequency bands, azimuths, and incidence angles, from the seismic swarm recorded in 2022 in the Tancitaro-Parícutin volcanic region. These features are used to train a multiblock convolutional neural network for seismic event detection and a physically oriented encoder-decoder model to identify the respective P- and S-wave phases of regional seismic events. An integrated deep learning model is built as a practical and automatic alternative to traditional seismic task solutions. Model performance and generalizability capabilities are tested via seismo-volcanic data from two stations in the Tacaná volcano region for November 2017.

Area of study

Mexico is strongly affected by the subduction process between the Cocos and North American plates. The preferential direction of the Cocos Plate relative to North America is N-NW, slightly clockwise from perpendicular to the trench, with rates of 5.5 cm/year for the Colima region, 6 cm/year in the Acapulco region, and 7.5 cm/year at the Chiapas-Guatemala border. Historically, Mexico's most significant seismic activity has occurred in the southern region, with relevance in Michoacán, Guerrero, and Oaxaca states. Significant examples include the 1978 San Pedro Pochutla, Oaxaca earthquake with $M_w=7.6$, the 1985 Michoacán earthquake with a magnitude of $M_s=7.5$, and the recent 2022 Coalcomán earthquake with $M_w=7.7$ (SSN catalog, 2024). Most focal mechanisms indicate relatively shallow thrust events attributed to subduction and, less commonly, more profound normal faulting events that could represent tensional deformation of the subducted plate.

The main geological feature resulting from the Cocos and North American interaction is the Trans-Mexican Volcanic Belt (TMVB). It consists primarily of andesitic to dacitic volcanic rocks from the late Miocene (approximately 11 million years ago). It comprises active volcanoes that traverse Mexico from Nayarit to Veracruz (Fig. 1). The volcanic rocks are associated with the subduction of the oceanic lithospheric Cocos Plate beneath the Rivera Plate. Compared with others the Trans-Mexican Volcanic Belt is an atypical continental arc, as it is farther from its forming trench, lacks defined continuity, and is wider than average. Furthermore, intermediate-depth earthquakes attributed to a subducted block beneath the Trans-Mexican Volcanic Belt have occurred. A study of this tectonic region has revealed a band of seismic foci submerged beneath southern Mexico at depths close to 100 km, defining what could be the Wadati-Benioff zone (Sedlock et al. 1993).

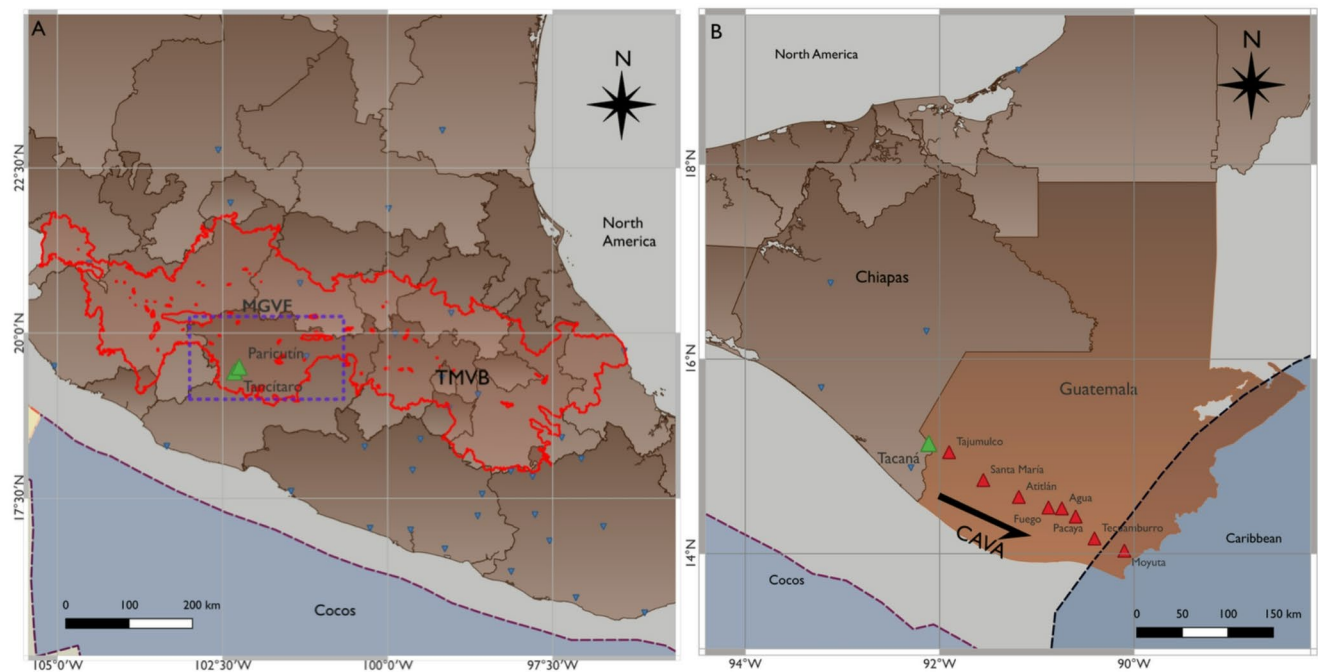


Fig. 1 Regional Tectonic and Volcanic Setting. **(A)** Generalized map of the western Mexican tectonic environment, highlighting the Trans-Mexican Volcanic Belt (TMVB) and the location of the Michoacán-Guanajuato Volcanic Field (MGVF), which hosts the Tancitaro and Parícutín volcanoes. **(B)** Tectonic and geographic context of the south-

ern region, illustrating the position of the Central American Volcanic Arc (CAVA). This arc initiates with the Tacaná volcano at the Mexico-Guatemala border and extends approximately ~1,300 km in a NW-SE direction toward Costa Rica

The Tancitaro-Parícutín Volcanic Region (TPVR) is situated in the western sector of the Trans-Mexican Volcanic Belt (TMVB), specifically within the state of Michoacán. This region is a component of the larger Michoacán-Guanajuato Volcanic Field (MGVF), which spans an area of approximately 34,500 km² and comprises an estimated ~1900 volcanic centers (Fig. 1.A). The geographical extent of the MGVF is delineated by -103° to -101° W longitude and 19° to 20° N latitude (Connor 1987). The dominant style of volcanism in the MGVF is monogenetic, characterized by the presence of cinder cones, lava domes, and maars. Additionally, the region contains approximately 400 medium-sized shield volcanoes that are larger than the typical cinder cones but smaller than the composite volcanoes prevalent in the TMVB. These shield volcanoes are primarily composed of lava flows, contrasting sharply with the pyroclastic nature of the smaller cinder cones (Hasenaka 1994). The MGVF's most prominent feature is the Tancitaro andesitic composite volcano. Rising 3,840 m above sea level, it stands more than 2,000 m above the surrounding topographical surface (Ownby et al. 2007). A relevant spatial distribution of monogenetic formations, including El Parícutín, El Jorullo, Astillero, El Pedregal, and El Jabalí, surrounds the Tancitaro volcano. Among these, the El Parícutín cinder cone is notable for its well-documented and complete eruptive history,

spanning from its onset in February 1943 until its cessation in March 1952. This event provided valuable insights into the monogenetic volcanic regime (Larrea et al. 2019). Recent scientific investigations focusing on the TPVR analyze the reactivation dynamics of this monogenetic complex, suggesting plausible trends toward renewed volcanic genesis. Specifically, Legrand et al. (2023) analyzed the occurrence of multiple seismic swarm events (1997, 1999, 2000, 2006, 2020, and 2021) within the TPVR, establishing a plausible preamble for monogenetic activity. Furthermore, the 2020 seismic swarm was comprehensively studied by Pertou et al. (2024), who detected approximately ~100,000 earthquakes. This extensive dataset delineated the tracks of magma migration and, in combination with the results from two seismic noise tomographies, led to the identification of two distinct magmatic reservoirs.

The Tacaná volcano is situated at the northern termination of the Central American Volcanic Arc (CAVA), an active segment that spans approximately 1,300 km from the Mexico-Guatemala border southward to Costa Rica (Fig. 1.B). The CAVA's genesis is fundamentally related to the complex triple junction involving the Cocos, Caribbean, and North America tectonic plates. This integrated relationship drives significant regional geological activity, which manifests as faulting, elevated seismicity, volcanism, and

regional (García-Palomo et al. 2006; Guzmán-Speziale et al. 1989). The Tacaná volcanic complex comprises three andesitic volcanoes and one lava dome, distributed between the state of Chiapas in Mexico and the San Marcos Department of Guatemala: Chichuj, Tacaná, the Las Ardillas dome, and Tajumulco. Reaching an elevation of 4,060 m above sea level (m.a.s.l.), the Tacaná volcano is the second most significant topographical feature in Central America, surpassed only by the Tajumulco volcano (4,220 m.a.s.l.) (Legrand et al. 2025).

Data

This section details the datasets utilized for this study, which include both primary data for training and external data for model generalization assessment. The seismic event detection and phase-picking models were primarily trained and tested using a seismo-volcanic dataset acquired from a broadband station operated by the National Seismological Service of Mexico (SSN) within the TPVR. This dataset comprises three-component seismic traces captured during the 2022 volcanic swarm. Subsequently, following a rigorous data science workflow, model performance is validated using external generalization data: the Tacaná 2017 dataset. This comparison is geophysically motivated, as the coherence in regional tectonic settings—which influences the physically driven feature scheme—permits a meaningful comparison. However, the distinct local volcanic mechanisms must be considered with caution during the final model evaluation.

Seismic event detection data

For the seismic event detection case, we used continuous waveforms from the National Seismological Service (SSN) for analysis. Specifically, we used three-component signals recorded from September to December 2022 at the broadband station TMIG, which is part of the recently deployed network surrounding the Tancítaro-Parícutín Volcanic Region (TPVR). The station is located at UTM coordinates $X=150\,412$ m, $Y=2\,142\,768$ m, $Z=2322$ m (Fig. 2). This dataset is particularly significant as small seismo-volcanic activity is suspected to have been triggered following the September 19th, 2022, Coalcomán earthquake ($M_w=6.8$). Additionally, the regional seismic rate increased considerably during this period (Legrand et al. 2023).

In a typical statistical learning workflow, the available dataset is divided into a training set and a test set, usually in proportions such as 75%/25% or 80%/20% of the total examples. This ensures that the model is exposed to enough examples during training to learn inherent features and patterns in the data. To avoid introducing bias, this division is performed randomly. To capture the diversity of seismic events, we included examples of different earthquake types (e.g., local and regional) in the training dataset. However, because the duration of events varies, we resampled them to standardize their length to 200-time samples (see Table 1). This resampling ensured consistency across the dataset and improved its suitability for model training.

The dataset comprises 6,680 examples, including seismic signal energy envelopes and noise samples. The energy envelopes were calculated from the three-component

Fig. 2 Seismic station array surrounding the Tancítaro-Parícutín Volcanic Region

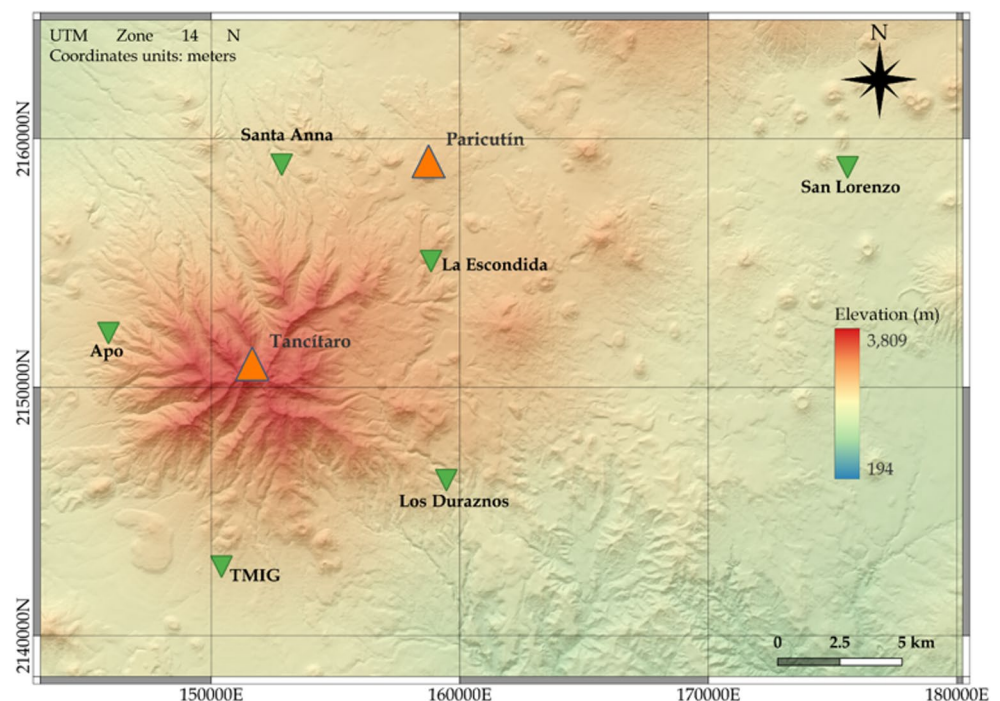
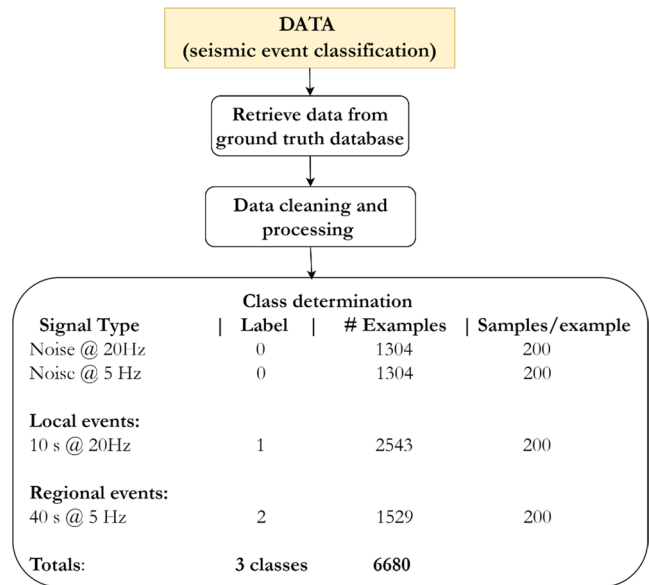


Table 1 Model training evaluation metrics for the local and regional seismic event detection classifier

Metric	Local detection	Regional detection
Loss	0.072	0.119
Accuracy	98.24%	96.875%
MCC	0.9643	0.9335
Training time	133 s	68.7 s

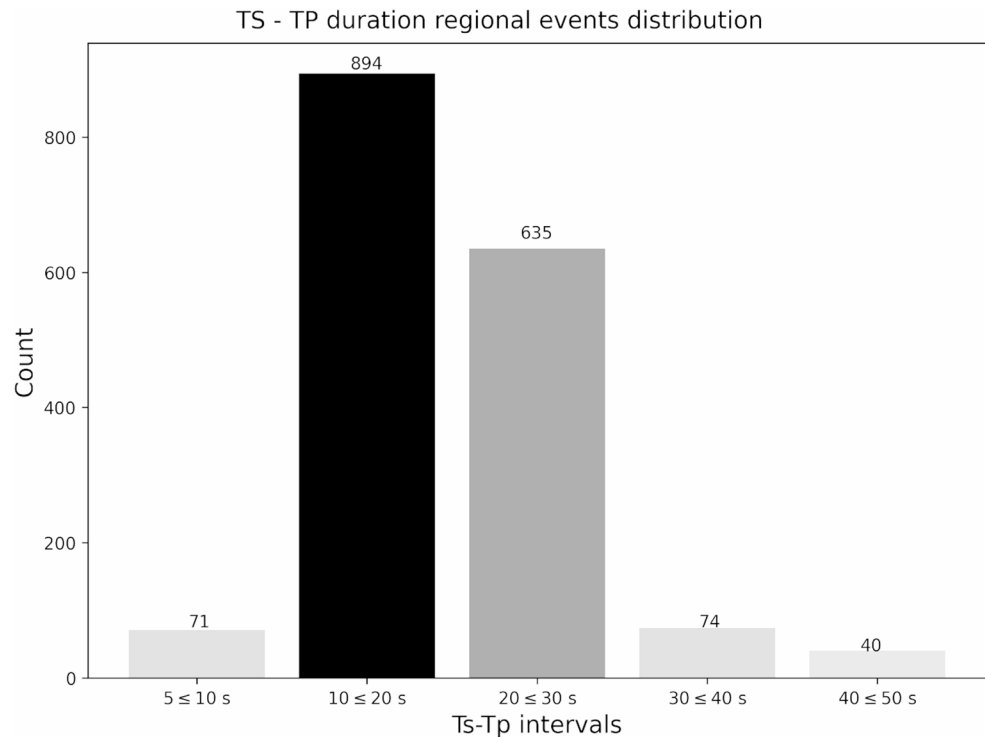
seismic signals recorded at the TMIG station between September and December 2022. The dataset includes 1,589 regional events sampled at 5 Hz (40-second event duration) and 2,543 local events sampled at 20 Hz (10-second windows). The variation in sampling rates was necessary to standardize all the samples to 200-time steps, regardless of the original duration of the seismic events. Additionally, 1,304 examples of ambient noise signals were included, with half sampled at 20 Hz and half at 5 Hz, resulting in a total of 2,608 noise examples (Fig. 3). This classification is based on a prior statistical analysis of the input database.

For regional events, we analyzed the differences between the P- and S-wave arrival times, as shown in Fig. 4. Five significant time intervals were identified: 5–10 s, 10–20 s, 20–30 s, 30–40 s, and 40–50 s. These intervals reflect the diversity of regional seismic events originating from multiple sources, as the arrival time differences serve as indicators source-receiver distance. Events with wave arrival differences between 10 and 30 s are the most representative in this dataset, with a total of 1,529 examples. However, events with differences of 30–50 s, corresponding to the farthest sources, are excluded from the training dataset because

**Fig. 4** Distribution of the data classes based on P- and S- time arrival differences

of their limited representation (114 examples across both classes). Similarly, events with differences of 5–10 s are excluded because their short time intervals are easily confused with local events, and the low number of examples is insufficient to represent this class adequately.

Seismic event discrimination from noise is treated as a binary classification problem. Seismic events are labeled with integer scalar values: 1 for seismic events and noise

Fig. 3 Diagram of the input database structure

signals are labeled 0. The model's objective is to classify seismic records with high accuracy into these categories using the physical feature of energy signal.

A multiblock convolutional neural network was trained to evaluate the classifier's sensitivity in distinguishing seismic events from noise. For local event examples, a 20-second window was used to capture sufficient seismic information, considering the in situ volcanic activity. For regional signals, windows of 40 s were selected to ensure enough samples to observe P- and S-wave arrivals and the decay of S-wave coda energy. Given that the average source locations along the Cocos subduction zone are 300–400 km from the TMIG station, the S-minus-P wave arrival times typically range between 30 and 50 s (Fig. 5). This windowing approach ensures the inclusion of essential seismic features for accurate classification.

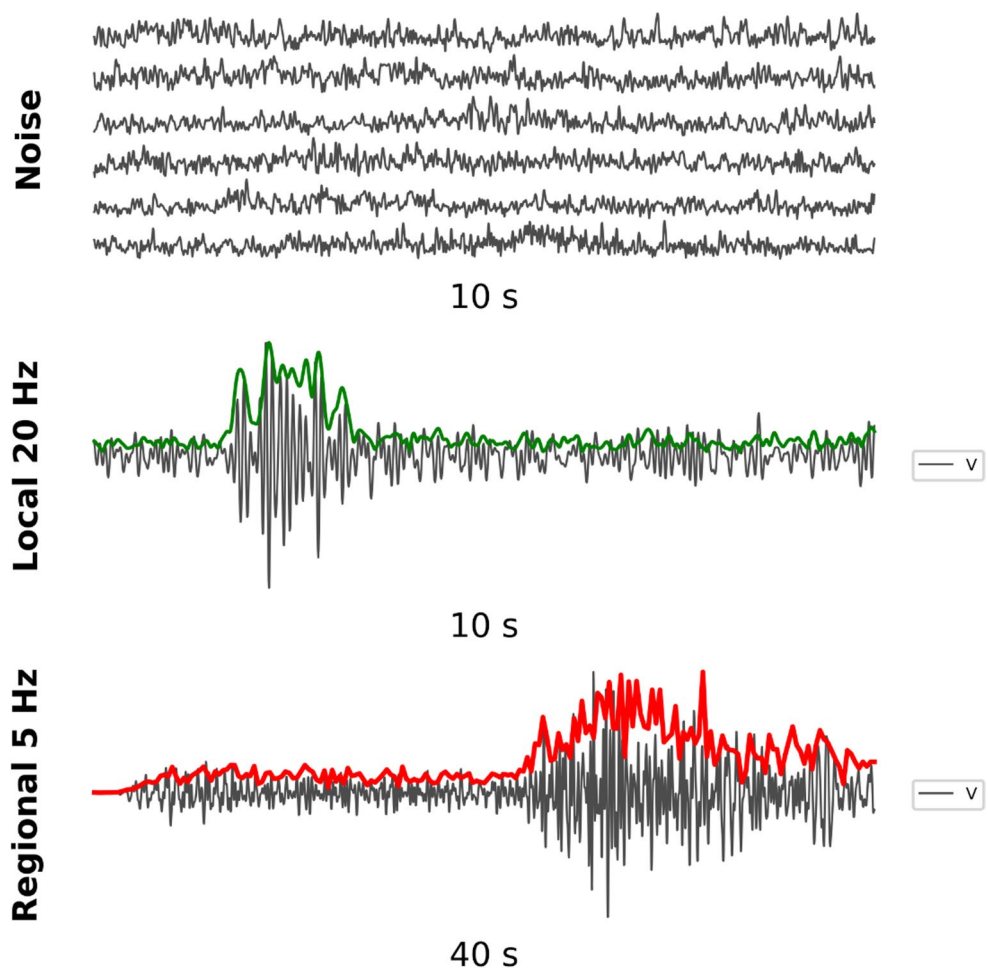
Additionally, the average seismic source generation along the Cocos subduction zone is approximately 300–400 km from the TMIG station. This distance corresponds to S minus P-wave arrival times ranging between 30 and 50 s, as previously indicated.

Phase picking data

The complete data set consists of 1,394 signals from regional events sampled at 100 Hz, with ground truth P- and S-wave time arrivals picked via the STA/LTA algorithm. This automatic algorithm was improved to remove biases related with the seismo-volcanic regime such as earthquakes overlap throughout the swarms, where noise sources add energy to the seismic signals resulting in altered P- and S-waves arrival times, especially for small earthquakes. The correction relies on testing other local energy maxima and comparing with the already picked signals, verifying if the waves time arrival differences are similar to the mean of the time arrival differences of all the signals, adjust better (Pertion et al. 2024).

The time arrival differences between the S-wave and P-wave arrivals range from 20 to 40 s. For the configuration of the raw signal database, windows of 50 s were considered, resulting in 3-component time series with 5,000 samples per example. The labels are generated by two functions that determine the probability of each time sample belonging

Fig. 5 Different types of seismic signals used in the seismic event detection case. The green and red signatures show the signal energy envelope of the original 3-component seismic response, for local and regional events, respectively



to either the noise or P-S-wave time arrival classes. These masks are Gaussian functions centered at the previously picked times using the ground truth arrival times for P- and S- waves. The Gaussian bell width is defined by a standard deviation adjusted empirically to retrieve significant samples before and after the determined time arrival sample and to minimize the incorporation of noise samples into the interest class labeling (Fig. 6).

The later consideration makes this classification problem the use of an imbalanced dataset, where the number of noise samples significantly exceeds the number of time arrival samples in time. For an example composed of 5000-time samples, the distribution of noise labels consists of 4,849 samples (~97%), while the number of time arrival samples is only 151 (Fig. 7).

Generalization test data

The data used to evaluate the generalization stage involve seismo-volcanic 3-component seismic signals recorded in the vicinity of Tacaná Volcano, located in Chiapas, Mexico. Compared to the 100 Hz sampling rate from the Tancitaro-Parícutín data, the measurements from the Tacaná stations have a sampling rate of 40 Hz. This results in a 60% loss of information which must be considered when assessing the model's generalizability capabilities. Initially, data from three stations were available: CHIQ, PAVE, and PATR. However, data from the PATR station contained significant noise that obscured the seismic signals of interest, so we excluded this station from the analysis. An example of a 1-hour seismic record from the CHIQ and PAVE stations is shown in Fig. 8, with the components superimposed on the signal energy envelope (represented by the magenta signal).

Fig. 6 Example of regional 3-component seismic event raw waveform

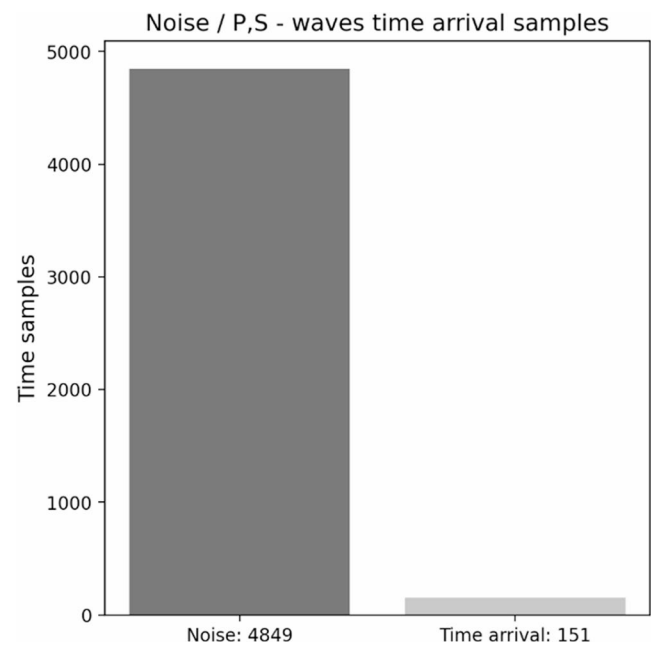
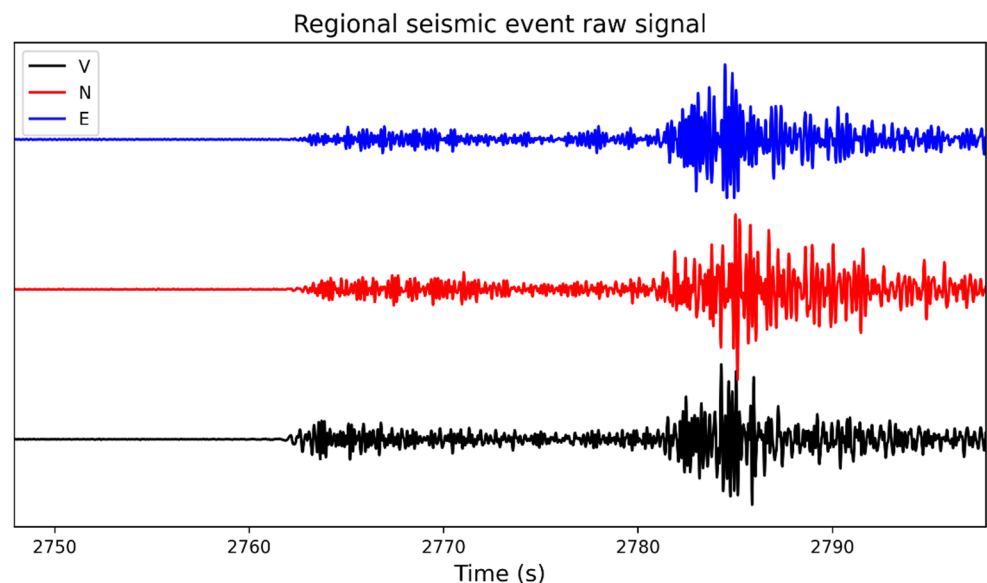
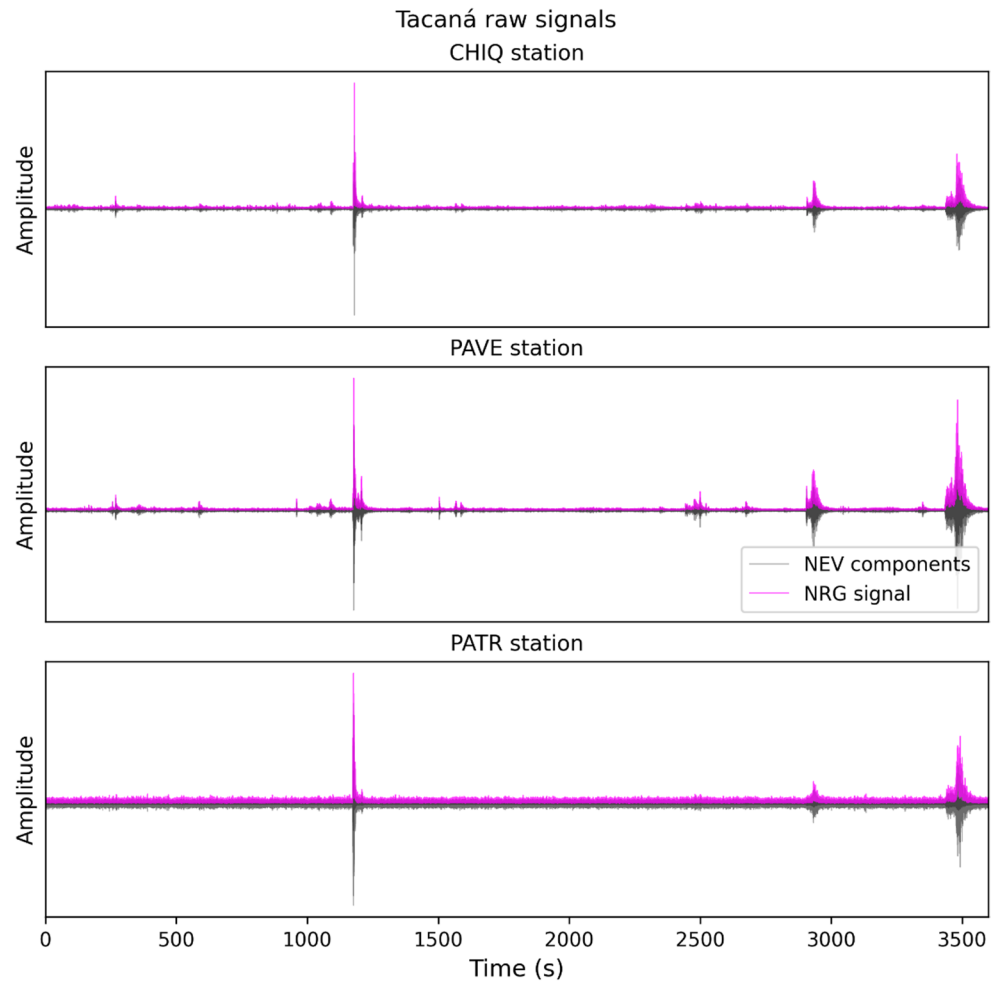


Fig. 7 Distribution of time samples in the noise and wave time arrival classes

The generalization process involves reading the database and extracting 1-hour signals from the CHIQ and PAVE stations. Windows are selected and overlapped by 50% along the signal. The duration of these windows is 40 s for regional detection and 10 s for local detection. For the regional detection case, the windows are resampled from 40 Hz to 5 Hz to achieve the 200-time samples required by the convolutional multiblock detection model. When a regional event is detected, the corresponding window is excluded from further analysis to reduce the chances of false detection by the local event detector. This procedure is sensitive to the initial time position of the windows.

Fig. 8 1-hour Tacaná raw seismic signals from CHIQ, PAVE, and PATR stations



Methodology

In what follows, we describe by detail the Integrated Deep Learning Workflow for Seismic Event Detection and Phase Picking. This integrated deep learning approach combines seismic event detection and phase-picking into a practical, efficient, and accurate framework that leverages physically driven seismic feature engineering (Fig. 9). The process is implemented in two primary stages: initial regional detection followed by phase picking and refined local detection.

Stage 1: Energy signal envelope and regional event detection and candidate screening (blue rectangle in Fig. 9).

The workflow begins by retrieving the raw three-component seismic data. To prepare the input for the preliminary classification, the energy signal envelope as the physical feature for this stage is computed at the original frequency sampling rate 100 (Hz).

To ensure consistent input tensor dimensionality (200-time samples), the data is resampled:

1. **Regional Events:** Resampled to 5 (Hz) for 40-second windows.
2. **Local Events:** Resampled to 20 (Hz) for 10-second windows (detailed in Stage 2).

The resampled data is then processed into 40-second windows with a 50% overlap applied along the entire signal trace. These windows serve as input to the regional-oriented 1D multiblock Convolutional Neural Network (CNN) classifier.

Any window with a model prediction of 1 is classified as a regional event candidate. Following this classification, the initial and final times of these candidate windows are retrieved, and the corresponding signal portions are removed from the original trace.

Stage 2: Feature Engineering and Phase Picking (magenta rectangle in Fig. 9).

For all windows effectively predicted as regional events (prediction=1), the original three-component signal portions are retrieved. If the original signal's frequency sampling rate differs from 100 (Hz), it is resampled to a consistent 100 (Hz).

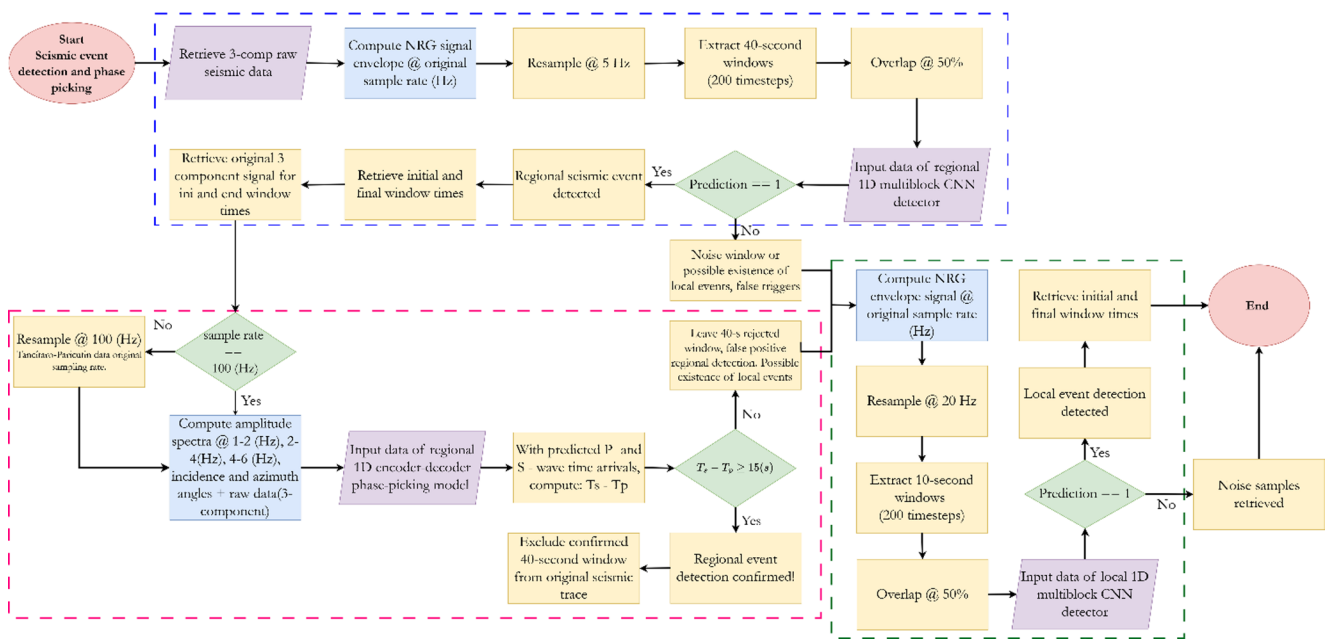


Fig. 9 Workflow for the Physically Driven Deep Learning Seismic Event Classification and Phase-Picking Joint Model. The diagram illustrates a two-stage detection framework that utilizes physically engineered features for enhanced event discrimination. The blue rectangle defines the initial regional seismic event detection process and outputs candidate events. Subsequently, the magenta rectangle

Crucially, physically driven features are then computed to provide comprehensive input for the phase-picking model. These features include:

- Amplitude spectra across specific frequency bands: 1–2 (Hz), 2–4 (Hz), and 4–6 (Hz).
- Incidence and azimuth angles.

The complete input, consisting of the original raw data alongside these physically driven features, is fed into a 1D encoder-decoder model for joint seismic event detection and phase picking.

The model analyzes the P and S-wave time arrival differences ($T_s - T_p$) to confirm the event type:

- If $T_s - T_p \geq 15$ (s), the regional event detection is confirmed, and the 40-second window samples are removed from the original trace.
- If $T_s - T_p < 15$ (s), the samples are retained, as this suggests a possible false positive regional prediction and a probable existence of a local event within the window.

Stage 3: Local Event Refinement (green rectangle in Fig. 9).

The final stage addresses windows rejected by the regional model. This includes samples where the initial window prediction was 0 (classified as noise but potentially containing

facilitates the phase-picking process, which is applied exclusively to the previously classified regional events. Finally, the green rectangle incorporates a specialized local seismic detection module, which is applied only to the rejected regional false positives and the regional true negative detections to ensure maximum completeness

a local event), or where $T_s - T_p < 15$ (s). These remaining windows are prepared for local event identification:

The energy signal envelope is computed, and the data is resampled at 20 (Hz). Shorter 10-second windows (200-time samples) are then extracted with 50% overlap. This processed input is passed to the 1D multiblock CNN local detection model.

- If the local model prediction is 1, a local event is determined, and its initial and final window times are retrieved.
- If the prediction is 0, the samples are definitively classified as noise.

This joint process ensures that both regional and local events are identified and accurately timed, demonstrating a practical, efficient, and robust application of a physically driven deep learning approach in seismo-volcanic monitoring.

Deep learning model: multiblock 1D-convolutional neural network for seismic event detection

The selection of a relatively simple multiblock 1D-Convolutional Neural Network (CNN) for seismic event classification is based on a structured set of assumptions designed to ensure computational efficiency, simplicity, and performance parity with established methods.

Strategic Integration of Feature Engineering: seismic event detection is a well-defined task traditionally addressed by algorithms that rely heavily on signal characteristics, such as the Short-Term Average/Long-Term Average (STA/LTA) ratio. Our proposed scheme is designed to leverage this established foundation by implementing a deep learning model that processes physically engineered signal attributes. The primary objective is to maintain the classification model as computationally simple as possible, thereby achieving an economic, reliable, automatic, and straightforward implementation that remains competitive with traditional methods, while addressing the problem of false detections caused by the complexity of the local-scale volcanic environments.

Leveraging the Energy Signal Envelope: the core physical feature employed is the energy signal envelope (NRG), derived from the raw three-component seismic data. This physical feature captures the amplitude variations intrinsic to seismic events, which have proven effectiveness in traditional detection schemes. The choice of a 1D CNN is appropriate because the convolutional and pooling (downsampling) processes are highly suitable for analyzing sequential time-series data. Specifically, the hierarchical structure of the CNN is effective in retrieving relevant attributes at various sampling rates, which, in our case, are initially induced by the computation of the energy signature from the three original signal components. This architectural choice therefore facilitates the optimal processing of the pre-processed, physically motivated input features.

This architecture consists of three blocks, each comprising convolutional, pooling, and dropout layers. In the convolutional layers, feature maps are generated by convolving optimized filters (kernels) with all one-dimensional input examples. These feature maps progressively increase as the training process refines the filters. The convolutional layer is the cornerstone of this model, and careful tuning of its hyperparameters is critical to achieve satisfactory results. The key hyperparameters include the number of filters, kernel size, and the activation function.

The pooling layer reduces the size of each example by subsampling, effectively halving the original window size. This process is controlled by the hyperparameter pool size. Following this, the dropout layer randomly deactivates neural nodes and their interconnections to mitigate overfitting. The drop rate hyperparameter specifies the percentage of nodes to deactivate (Fig. 10).

After these three blocks, a flattening layer converts the 3-dimensional structure of the processed data into a 2-dimensional array, which prepares it for input into the dense layers. A dense layer with 64 neural units follows, applying a progressive reduction in dimensions inherited from the preceding convolutional process. The output of this dense layer passes through a final dropout process to further prevent overfitting.

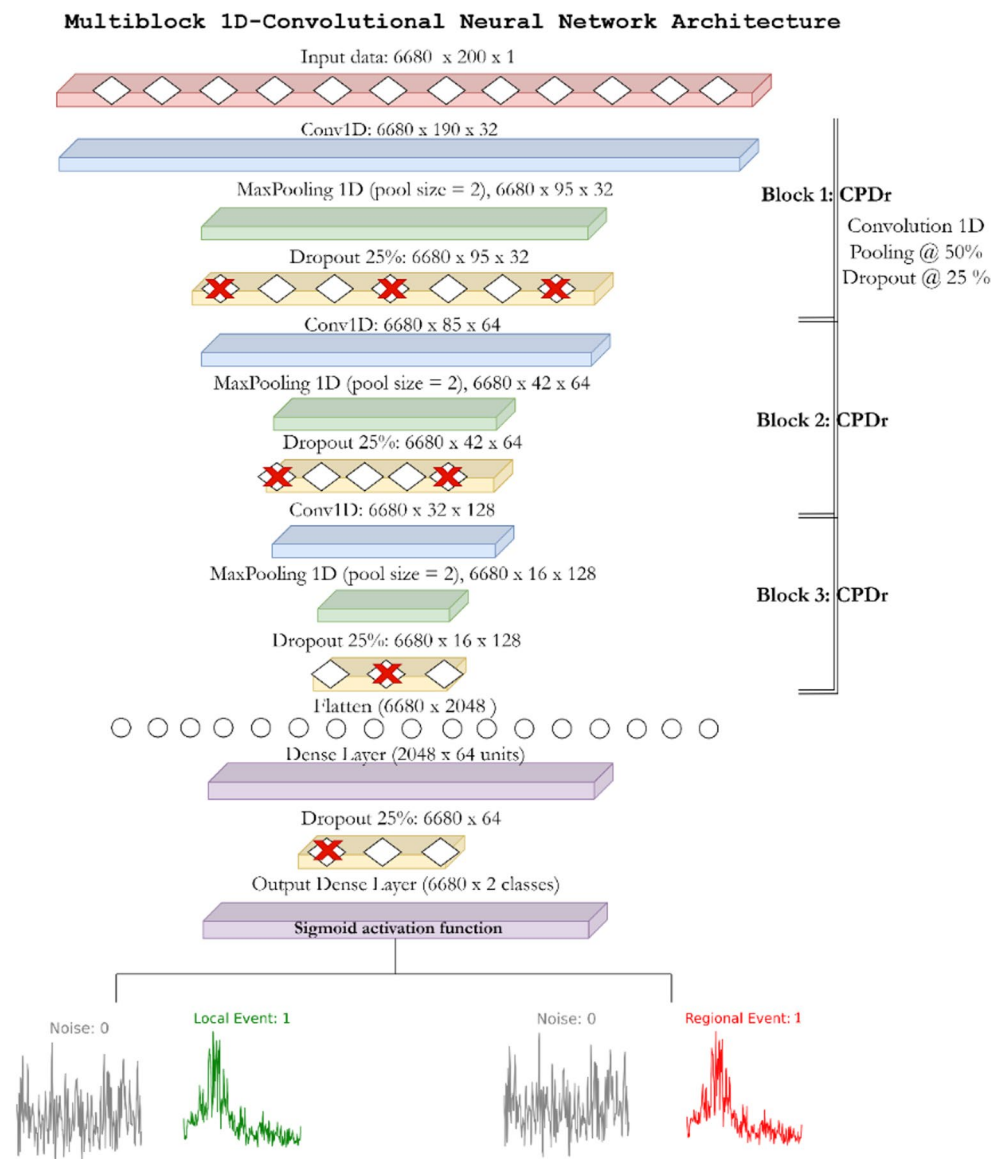
Finally, the results are fed into the output dense layer, which calculates the probability of each example belonging to a specific class (e.g., event or noise). This probability is computed via the activation function in the output layer. For this binary classification problem, the sigmoid or logistic function is employed. All the implementations are coded in Python, and for the deep learning model development the *tensorflow* API was used (Martín et al. 2015).

An encoder-decoder neural network for seismic phase-picking

The model used for phase-picking implementation is based on an image processing technique known as *image segmentation*. This method focuses on masking groups of pixels that form objects within the image. In this context, each group of pixels defines a distinct class. For example, in a landscape, the pixels representing the sky are considered a different class from those forming the mountains in the background, and similarly, the pixels corresponding to the grass at the bottom of the image form another class. One of the most promising deep learning models for solving segmentation tasks is the encoder-decoder architecture. The encoder part involves convolving the input data to extract patterns and features in the form of attribute maps, which are then used to calculate pixel-level class membership probabilities. Decoding involves deconvolution and resampling operations that help identify the classified pixel masks. Some deep learning models used for general image segmentation include DeConvNet (Noh et al. 2015), SegNet (Badrinarayanan et al. 2017), and HRNet (Yuan et al. 2020).

Medical imaging has significantly contributed to the implementation of deep learning methods, particularly in enhancing the accuracy and speed of early diagnoses. Semantic segmentation of medical data has enabled the early detection of tissues affected by cancer cells. A notable model in this field is U-Net (Ronneberger et al. 2015). U-Net consists of two primary structures: the first, known as the encoder, contracts the image through convolutions, creating feature maps (patterns recognized through the application of kernels, which are updated at each training epoch).

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Fig. 10 Multiblock 1D-convolutional neural network schematics

Common operations in convolutional neural networks, such as downsampling and probability determination via the ReLU activation function, are performed. Downsampling occurs at a ratio of two, meaning that the image is reduced to half its pixel count in each depth block of encoding. Conversely, the number of feature maps calculated in each block doubles. The expansive part of the network (the decoder) consists of upsampling operations, which double the number of pixels recovered by deconvolution of the attribute map and optimized filters. Unlike the first stage of the network, the attribute maps are halved, and concatenation with the corresponding attribute map from the encoding path is performed. Finally, the output is obtained through one-dimensional convolution to map the 64-component attribute vector to the number of classes of interest.

In this work, we adapted the U-Net architecture to analyze time series data rather than images. Since convolutions must be applied to 1-D data, the dimensions of the input array were modified accordingly. Additionally, some hyper-parameters, such as the kernel size and stride, were adjusted to fit the 1D nature of the time series. Five convolution-pooling blocks were applied to the encoder, whereas five deconvolution-upsampling blocks were required to recover the original sampling of the input signal and the analyzed feature maps. The output section comprises a convolution layer with a number of features equal to the number of classes. Finally, a time-distributed dense layer with a sigmoid activation function determines the probability of each sample belonging to a specific class. The proposed architecture consists of a total of 28 layers, representing over 16 million trainable parameters (Fig. 11).

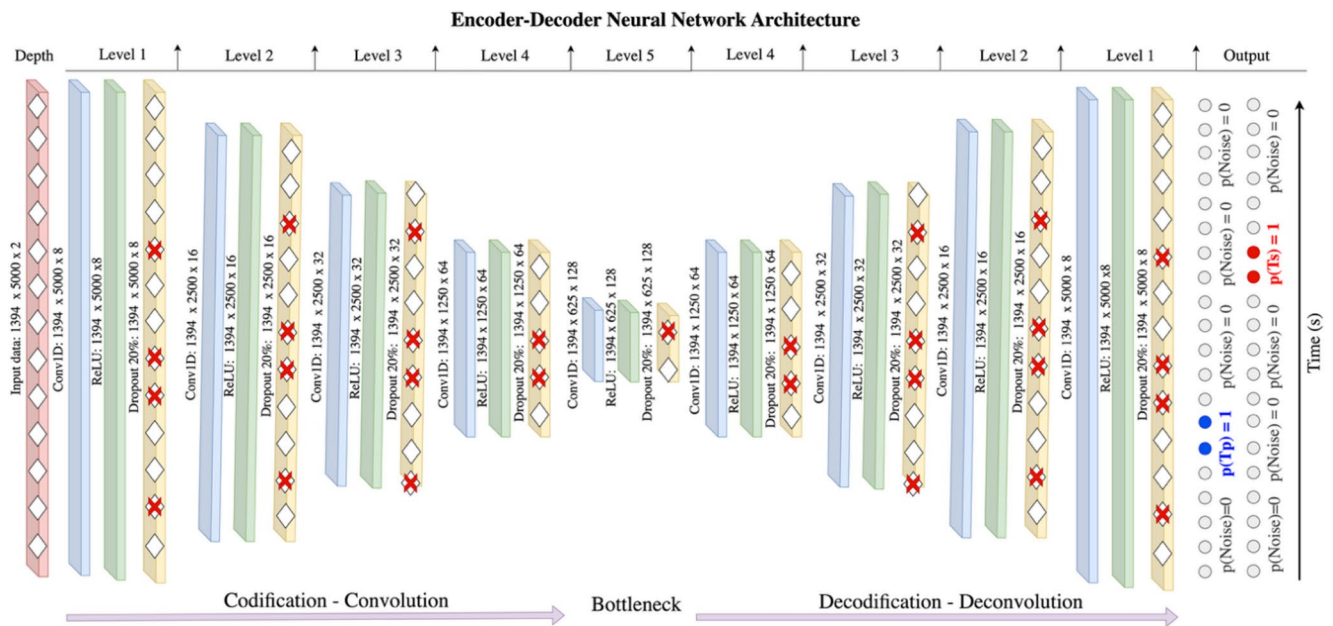


Fig. 11 1D encoder-decoder architecture

Physically driven feature engineering

The physically driven feature engineering proposed in this work applies standard transformations used in seismology, such as computing the energy envelope, amplitude spectra for multiple frequency bands, azimuths, and incidence angles. Accurate seismic event detection necessitates a reliable feature that efficiently encapsulates signal characteristics while minimizing computational overhead. The energy signal envelope fulfills this requirement, serving as a physically driven attribute that optimizes both data storage and training computational cost—a key factor in achieving our goal of developing an economic and reliable deep learning model competitive with established traditional algorithms.

The energy envelope, $\text{NRG}(t)$, represents the magnitude of the complex analytic signal, $H(t)$, which is computed directly from the three-component raw seismic trace. Mathematically, the analytic signal is expressed as:

$$H(t) = s(t) + i \cdot \text{Hilbert}(s(t)) \quad (1)$$

$$\text{NRG}(t) = \sqrt{\text{Re}(H(t))^2 + \text{Im}(H(t))^2} \quad (2)$$

where $s(t)$ is the original seismic trace. The real part of $H(t)$ corresponds to the original trace, and the imaginary part is the Hilbert transform of $s(t)$.

By capturing the amplitude modulation of the seismic signal, $E(t)$ effectively summarizes the overall energy distribution over time across all three components. This data

reduction step is critical, as it provides a robust, single-channel representation that retains the essential information required for accurate classification, thereby directly supporting the simpler architecture of the multiblock 1D-CNN model.

For the case of seismic phase-picking the amplitude $A_f(t)$ is computed from the three signal components $N_f(t)$, $E_f(t)$, and $V_f(t)$ at specific frequency intervals (f), as follows:

$$A_f(t) = \sqrt{N_f(t)^2 + E_f(t)^2 + V_f(t)^2} \quad (3)$$

These frequency bands are derived from a FTAN signal analysis. For these seismo-volcanic signals, the most relevant information lies within the 2–8 Hz frequency range. We chose to filter the signals within 1–2 Hz, 2–4 Hz, and 4–6 Hz intervals (Fig. 12).

The azimuth angle $Az(t)$ is an angle that comprises the horizontal components in the form:

$$Az(t) = \tan^{-1} \left(\frac{E(t)}{N(t)} \right) \quad (4)$$

The incidence angle (ϕ) considers the ratio between the horizontal components with respect to the vertical component (Fig. 13):

$$\phi(t) = \tan^{-1} \left(\frac{E(t)^2 + N(t)^2}{V(t)^2} \right) \quad (5)$$

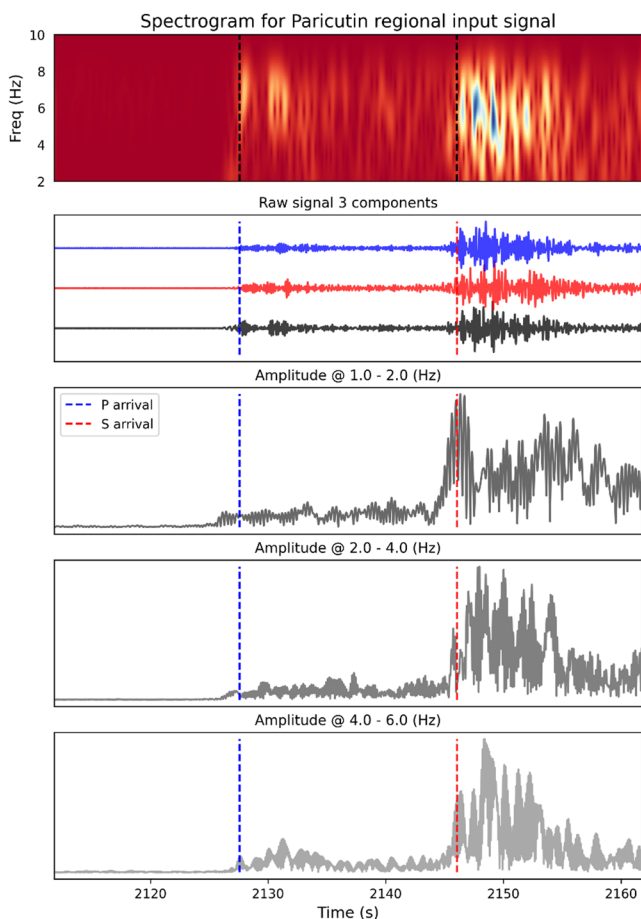


Fig. 12 Spectrogram and amplitude spectra for different frequency bands of a regional seismic event sample

Generalization test

Generalization procedures aim to demonstrate the effectiveness of deep learning models when analyzing new data specifically, external data that were not included in the training and testing datasets during the initial stages. This stage is often referred to as the application of the trained model, prior to its implementation in real-world problems.

Applying our trained model to a different volcanic configuration, such as the Tacaná-Paricutín Volcanic Region (TPVR), requires a rigorous generalization test. This procedure is a fundamental component of the established data science workflow, requiring analysis of the model's performance on data entirely exogenous to the original training and validation datasets. Specifically, the test is designed to assess the model's capacity to recognize characteristic seismic event patterns beyond the local geological and geophysical conditions of the TPVR.

The selection of the Tacaná dataset for this generalization test is motivated by a combination of regional tectonic similarity and data availability constraints:

1. **Tectonic context:** The Tacaná volcanic region is influenced by the subduction process of the Cocos Plate beneath the North America Plate, but it is also affected by secondary tectonic interactions involving the Caribbean Plate. This regional setting exhibits a complexity that is relevant to testing the model's robustness.
2. **Physical feature similarity:** A crucial justification for the test lies in the observed physical similarity between the Tacaná and TPVR seismic signals, particularly concerning regional events (as detailed in Sect. 5.3). Specifically, the two datasets exhibit comparable frequency ranges and similar amplitude responses, which suggests that the regional component of the seismic signatures is sufficiently consistent for cross-application. This physical concordance strongly supports the feasibility of using the Tacaná data to evaluate the model's generalization capabilities at a regional scale.

Furthermore, it is important to acknowledge the inherent limitations of this cross-regional test. Given the distinct in situ magmatic and hydrothermal processes governing the two volcanic systems, greater differences are expected at the local seismic scale. Consequently, the performance of the local event detection component may be biased by the unique characteristics embedded within the original TPVR training data. This constraint highlights that the generalization success is expected to be more robust for the regional event classification than for the specialized local event classification.

Results and discussion

Regional and local seismic event detection: training and testing on the TPVR dataset

After training simpler architectures with a single convolutional-pooling-dropout block were tested and suboptimal predictions were observed, a multiblock convolutional neural network was implemented. This setup demonstrated improved performance in distinguishing seismic events from noise under the proposed scheme.

Various frequency samplings of the signals were evaluated to identify the model that performed best in event classification. However, overfitting emerged as a significant challenge during training. While dropout operations were included to mitigate this issue, they were insufficient. To address overfitting further, regularization techniques, particularly L2 regularization, were explored. Different scalar factors for the L2 penalty were tested to identify the optimal configuration for effectively minimizing the loss function.

The black curves represent the test evaluations performed throughout the training process. The complexity of the

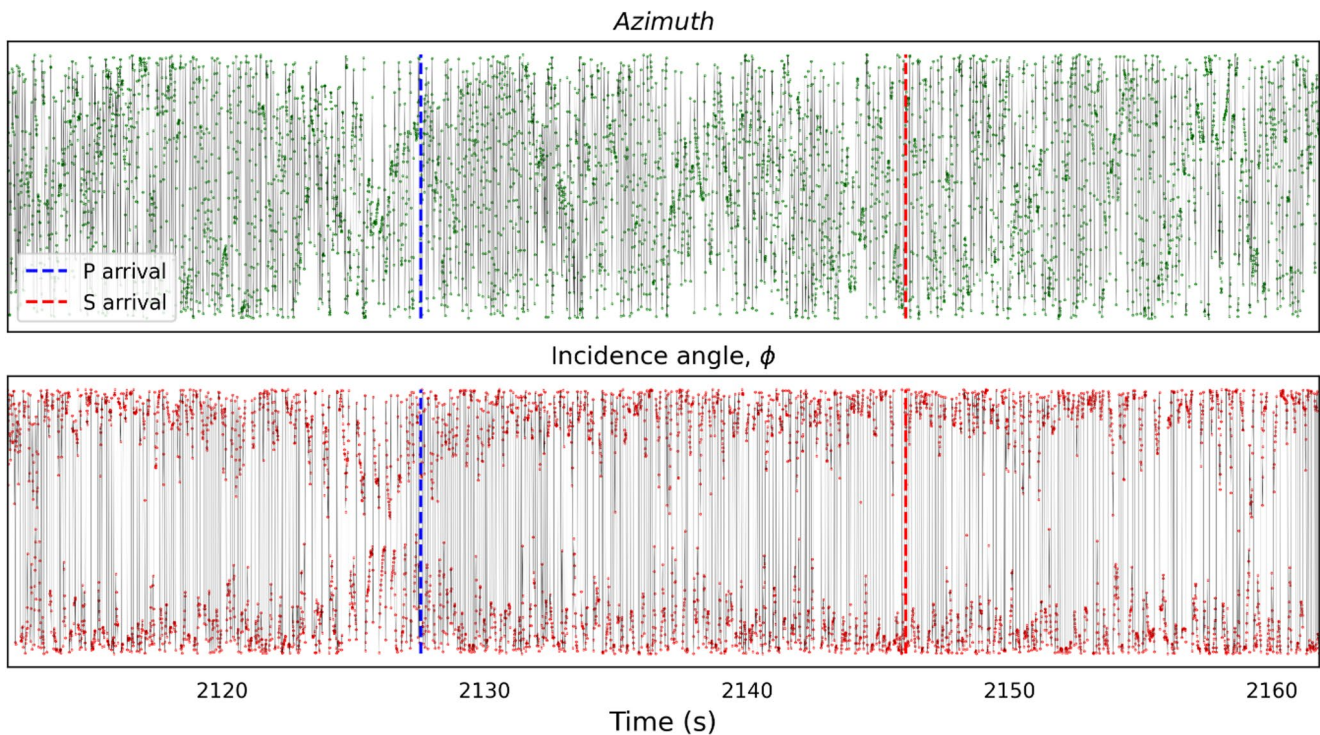


Fig. 13 Incidence and azimuth features computed from signal components

model tends to overfit the training data, which is one of the most common issues in deep learning models. To address this, we analyzed the learning curves of loss and accuracy (Fig. 14). During the initial training sessions, the model

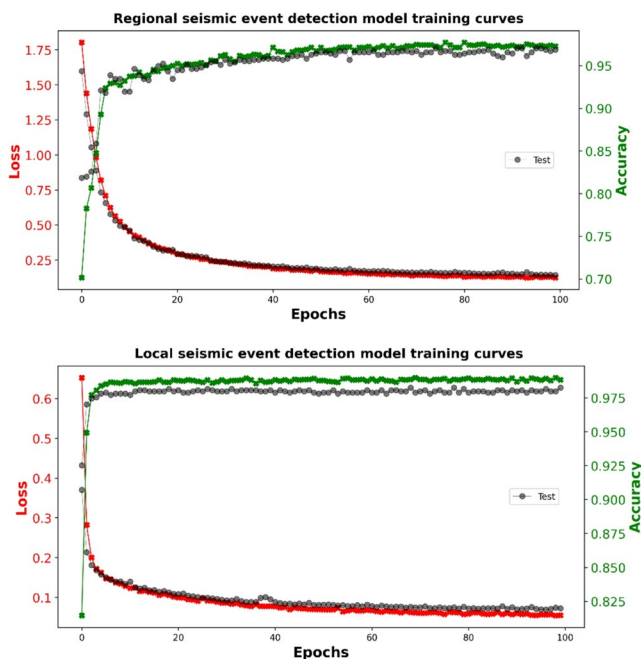


Fig. 14 Regional and local seismic event classification models training curves for loss and accuracy metric

began to overfit after 30 epochs, even with the application of dropout layers in the convolutional blocks.

To mitigate this issue, we implemented regularization techniques in both the convolutional layers and the dense layer before the flattening process. Using an L2 regularization approach with a penalty factor of 0.001 was enough to prevent overfitting. By increasing the number of epochs from 100 to 1000, we observed that overfitting did not reappear; however, the improvement in the model occurred at a slower rate. This led to the conclusion that training for 100 epochs with L2 regularization was sufficient to obtain reliable models.

As shown in Fig. 14, the loss and accuracy curves exhibit the expected behavior for a well-performing classifier: the loss gradually approaches zero, whereas the accuracy exceeds 0.95.

The evaluation performance for seismic event detection is measured via three key metrics: accuracy, loss, and Matthew's correlation coefficient (MCC). Accuracy (A) is defined as the ratio of correct predictions (true positives and true negatives) to the total number of predictions, including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This metric provides an overall assessment of the classifier's performance, which is especially suitable for balanced datasets.

$$A = \frac{TP}{TP + TN + FP + FN} \quad (6)$$

The loss metric quantifies the error at each training iteration. For this multiclassification problem, the categorical cross-entropy loss function is used: which is expressed as:

$$L(y) = -\frac{1}{N} \sum_i^N \sum_j^K (y_{ij} \log(\hat{y}_{ij})) \quad (7)$$

Here, y_{ij} represents the true labels, $\hat{y}_{ij}$ represents the predicted probabilities, and i and j are the indices for the samples and classes, respectively. The predictions for classification are computed with the sigmoid function, as described previously.

As reported by Chicco and Jurman, (2020), Matthew's Correlation Coefficient (MCC) is preferred over metrics like F1-score and accuracy, particularly for unbalanced datasets. MCC evaluates the quality of predictions by balancing true positives and true negatives, assigning a value between -1 and $+1$. The MCC formula is:

$$MCC = \frac{TP \bullet TN - FP \bullet FN}{\sqrt{(TP + FP) \bullet (TP + FN) \bullet (TN + FP) \bullet (TN + FN)}}$$

The final metric values for the classifier are summarized in Table 1.

In Fig. 15, we present the confusion matrices used to evaluate the performance of the classifier. Confusion matrices are graphical tools used to analyze the performance of deep learning models when applied to predictions on a test dataset. As a visual tool, it makes it easier to identify where the model struggles, particularly in terms of true positives (the main diagonal of the matrix), false positives, true negatives, and false negatives (the nondiagonal elements).

Once again, overall good performance is observed, as the highest values appear along the main diagonal, which represents the total number of true detections, whereas

the off-diagonal values are smaller. This finding indicates that the model effectively discriminates between regional and local seismic signals, showing an acceptable separation from background noise. The model evaluation scores demonstrate high performance in seismic event detection, with a final accuracy of 98.24% and 96.875%, and loss values of 0.072 and 0.119 for the local and regional models, respectively.

From the analysis of the regional confusion matrix, 21 false detections are identified. The number of noise examples misclassified as seismic signals is minimal (7, shown in the first row, second column of the confusion matrix), while the number of seismic signals misclassified as noise is also low, with 14 misclassifications (shown in the second row, first column of the confusion matrix). This finding demonstrates that the trained model is effective in separating noise from the target signals, with an error rate of 2.08% (Fig. 15A).

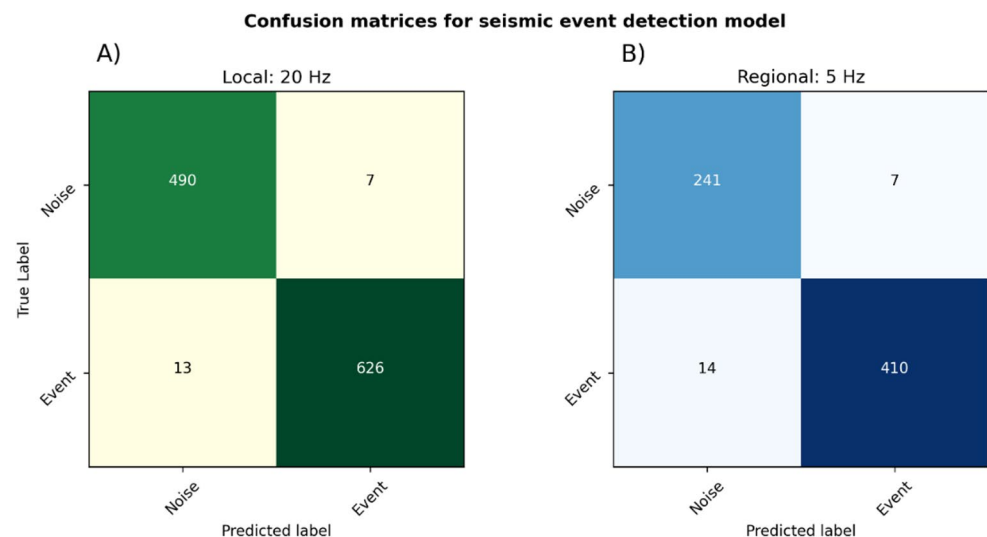
The model performs well on local event samples. Most local examples are correctly classified, with an accuracy rate of 98.24%. There are 20 misclassified examples, where 13 seismic signals are classified as noise, and 7 noise samples are predicted as events. The error rate for the local seismic event classification is 1.76%.

The MCC for the regional event classification problem is 0.9335, and that for the local model is 0.9643, which indicates a good- to-excellent score for the model test results in both cases.

Regional phase picking results: training and testing on the TPVR dataset

The problem in the phase-picking case involves the classification of several time samples, resulting in an imbalanced dataset. This type of input data does not have an

Fig. 15 Confusion matrices for the local and regional seismic event detection



equitable distribution of labeled examples, as each time sample belongs alternatively to the noise, P-wave, or S-wave time arrival classes. Noise samples are the dominant class, whereas only a few samples in each window correspond to P- or S-wave time arrivals. Since this is a binary classification problem, the corresponding loss function is computed via the binary cross-entropy function, which calculates the negative logarithm of the difference between the current prediction and the original label by comparing two probability distributions. This loss function is computed as:

$$L(y) = -\frac{1}{N} \sum_{i=1}^N y_i \bullet \log(p(\hat{y}_i)) + (1 - y_i) \bullet \log(1 - p(\hat{y}_i)) \quad (9)$$

where y_i is the true label, and $p(\hat{y}_i)$ is the probability determined by the classifier.

The evaluation of classification problems on imbalanced datasets is typically conducted via various statistical metrics, such as precision, recall, the F1-score, and Matthew's correlation coefficient (MCC). Metrics such as accuracy are not well-suited for imbalanced datasets and can be misleading in the interpretation of results, so they are omitted in the current analysis (Chicco and Jurman, 2020).

Precision is the ratio of true positives to the total predictions made by the model, showing the classifier's ability to avoid labeling false samples as true. It is computed via the following formula:

$$P = \frac{TP}{TP + FP} \quad (10)$$

Recall, also called sensitivity, is the ratio of true positives to the sum of true positives and false negatives. It demonstrates the model's ability to identify all the positive samples:

$$R = \frac{TP}{TP + FN} \quad (11)$$

The F1-score is the weighted harmonic average of precision and recall

$$F1 = \frac{2PR}{P + R} \quad (12)$$

Good F1-scores tend towards 1, whereas poor scores approach 0. To evaluate the performance of the feature engineering application, the model was trained using different feature combinations. The reference model was trained using the raw input dataset (3-component seismic series) only. The model's hyperparameters remained the same for all executions (Table 2).

Table 2 Phase-picking classifier hyperparameters

Hyperparameter	Value
Convolution-deconvolution depth	5
Kernel size	3
Pooling size	2
Dropout	20%
Stride size	2
Learning rate	0.001
Initial number of convolutional features	64
Number of epochs	19

Table 3 shows the model evaluation results. An overall improvement in performance occurs as more features are added during training. The model trained with the original data returns an F1-score of 19.16%, whereas the model trained using both incidence and amplitude features achieves a value of 66.50%, representing nearly a 50% improvement without significantly increasing the computation time during the training phase. The input data size tensor increased from 3 components in the third dimension to 7 components.

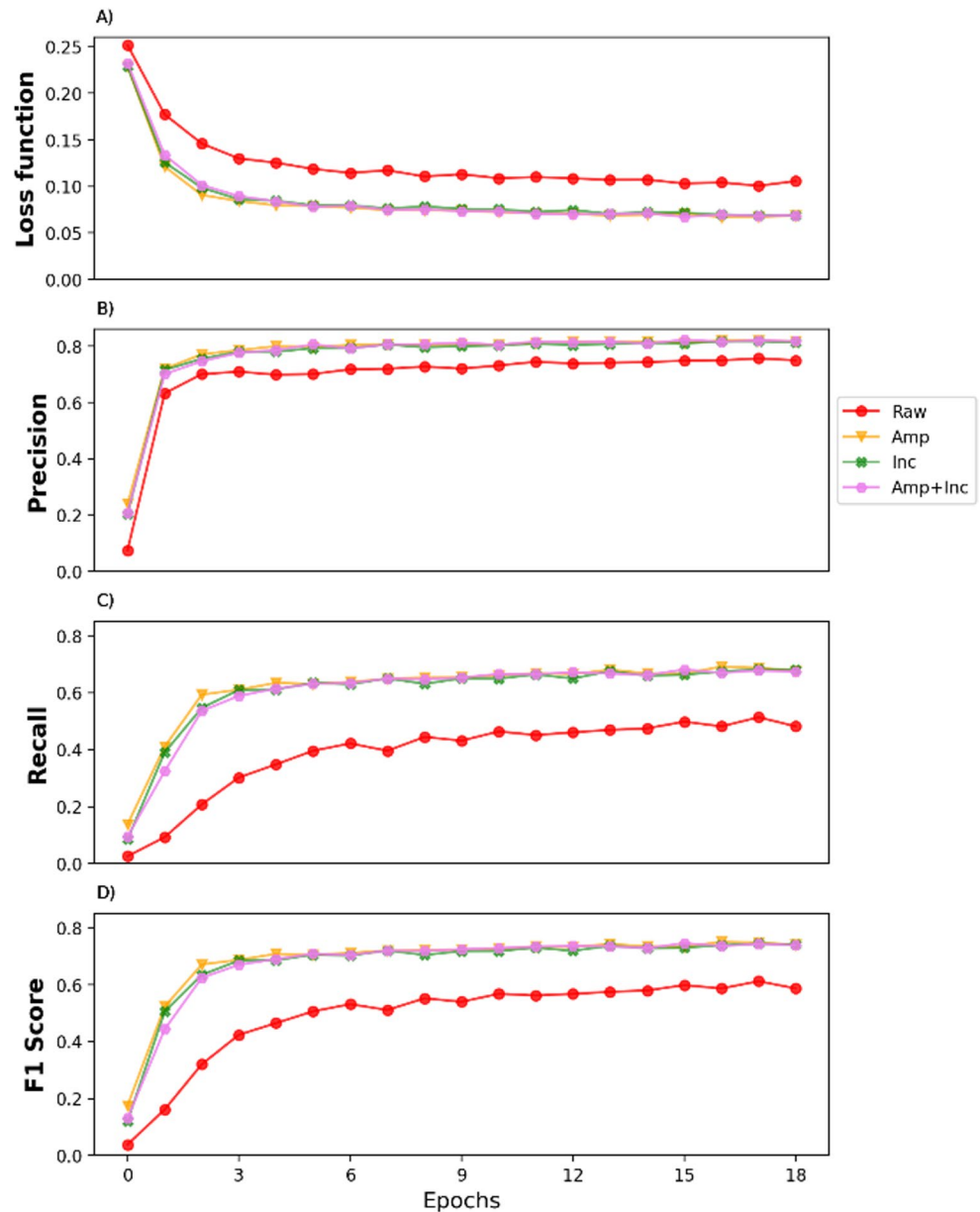
Compared with the use of individual features or combinations of features, raw model training results in lower performance throughout the entire training process. The use of the amplitude alone yields overall good performance, but the highest F1-score is achieved when the amplitude and incidence angle features are used together. This combination results in a 47.34% improvement over the use of raw data alone. The recall metric also shows a significant improvement, increasing by 43.43%. This suggests a strong relationship between the improvement in sensitivity for predicting true positive samples and the accurate determination of S-wave phases (Fig. 16).

The Matthew correlation coefficient (MCC) is particularly effective for evaluating deep learning models performance on imbalanced datasets. For this binary classification problem, we propose a graphical tool for assessing predictive performance based on the MCC metric rather than relying on the confusion matrix, which is not suitable for a problem of this nature. A classification problem is considered

Table 3 Model training evaluation metrics for different combinations of features

Metric/Model	Raw input	Only Incidence	Only Amplitude	Incidence + Amplitude
Loss	0.2335	0.1141	0.0754	0.0785
Precision	0.8209	0.9589	0.9014	0.8581
Recall	0.1085	0.2824	0.5094	0.5428
F1-Score	0.1916	0.4363	0.6510	0.6650
Computation time	30.296 min	32.116 min	32.283 min	31.331 min

Fig. 16 Training curves for different evaluation metrics



successful when predictions yield a positive MCC, with values approaching +1 indicating better performance. To facilitate visual interpretation, scatter plots were employed to represent MCC values.

The MCC scatter plots (Fig. 17) reveal improvements in identifying correct S-wave phases. Each point in the plots corresponds to the MCC value for one of the 273 test samples. Blue dots denote P-wave arrival time classifications, while red dots represent their corresponding S-wave arrival time classifications. The model's predictive quality is reflected in the clustering of points near +1, indicating optimal MCC values. In Fig. 18A, red dots cluster near 0, with several extending into negative values, suggesting

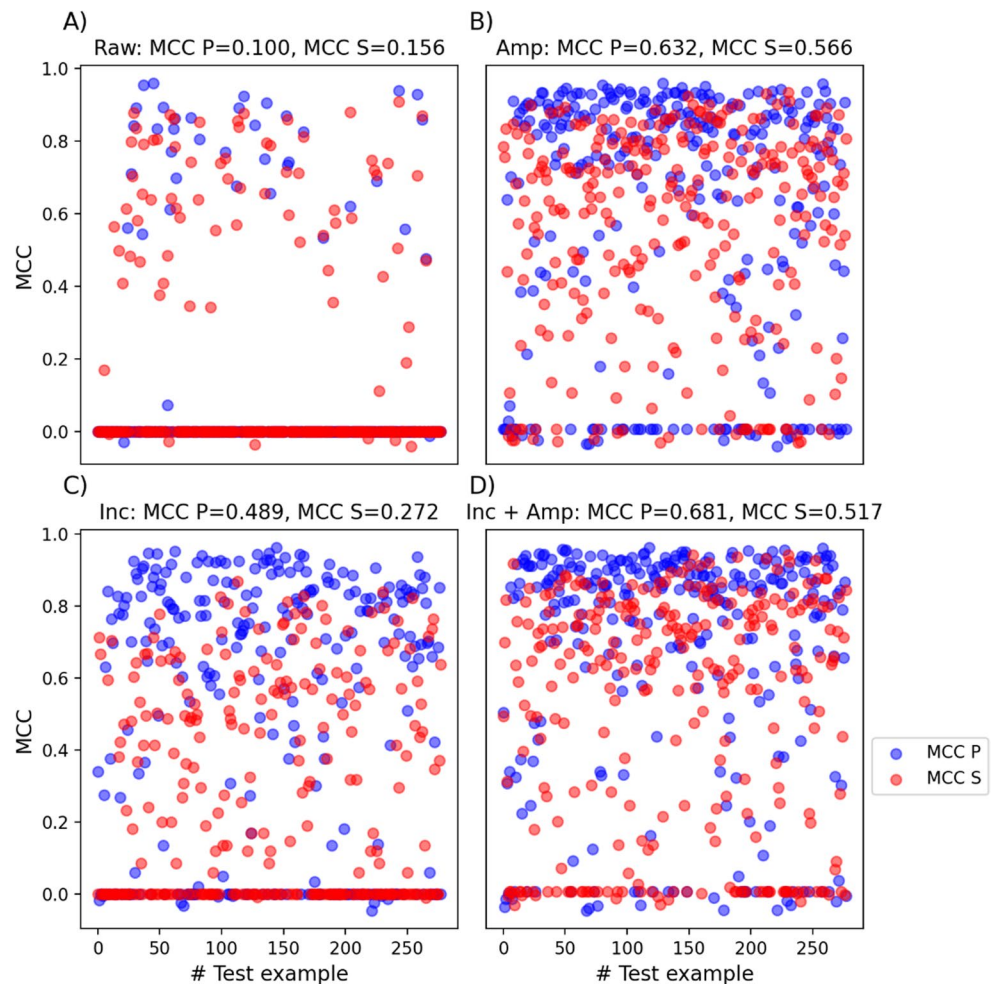
poor classification performance for S-wave arrivals using only the raw data model. In contrast, Fig. 18B demonstrates a significant improvement when the amplitude feature is incorporated.

Finally, 30 random test results were extracted from the test dataset and are shown on Fig. 19.

Generalization testing results on the Tacaná volcanic region dataset

In the example shown in Fig. 20, a regional seismic event is detected by the model in both stations, using 1 h of data from the observation on November 1st, 2017, starting at

Fig. 17 Scatterplot for Matthew's correlation coefficient in each feature combination case



00:00 (detections are highlighted in red). With 50% overlap, a total of 180 input examples is provided to the model. Some notable differences between the results can be observed. The detection model is sensitive to low-amplitude signals, as seen in the CHIQ signal between 1500 and 1650 s. These low-amplitude signatures are not present in the PAVE measurements. By using a normalized amplitude, the model is able to identify patterns related to the temporal variation in waveforms, independent of whether the amplitude is high or low. This feature enhances detection in high ambient noise environments. However, the model is also sensitive to false detections of signatures that do not have a seismic origin. Expert human supervision is always required to analyze the predictions made by the multiblock convolutional neural network.

Considering the seismic activity recorded by SSN stations in November 2017, the number of detected regional events is consistent with expected tectonic behavior. SSN registered 534 events with magnitudes ranging from Mw

3.0 to 7.0, all located in the vicinity of the Tacaná volcano (Fig. 21). The average magnitude and depth were 3.8 and 44.17 km, respectively. The CHIQ station exhibits greater sensitivity to noise compared to the PAVE station, resulting in a substantial number of detections associated with high-amplitude noise signals.

After detecting regional events, the convolutional model is applied to identify local seismic events. Special attention must be paid to false detections, as many high-amplitude pulses are embedded within regional events; therefore, thorough window removal is essential to reduce the occurrence of false triggers. The results of total event detections are summarized in Table 4. Model predictions are compared with manual detections from prior work. At the CHIQ station, 20 out of 22 local seismic events are confirmed, while at the PAVE station, 21 out of 22 are correctly identified by the convolutional detector (Fig. 18). The model produced 241 false detections at CHIQ and 286 at PAVE. Given the model's 98% accuracy on test data, these false predictions

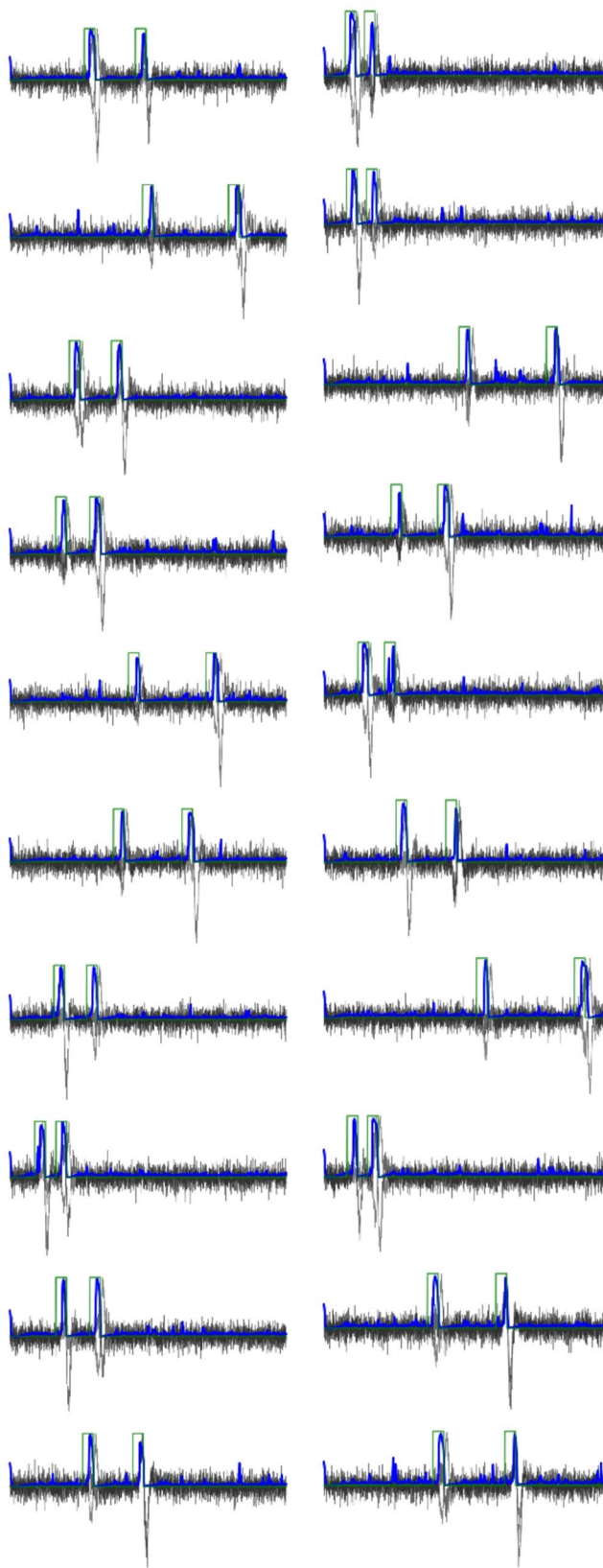


Fig. 18 Local (blue shade) and regional (red shade) detections for Tacaná signals

fall within the expected error margins, considering a total of 282,713 and 432,662 10-second windows for CHIQ and PAVE stations, respectively.

The phase-picking generalization test was applied to the previously detected regional seismic events from the CHIQ and PAVE stations. Importantly, the original data from Tacaná have a sampling rate of 40 Hz, which differs from the 100 Hz sampling rate used during encoder-decoder model training. The first stage involves adapting the new data to the appropriate settings required for this deep learning algorithm.

Owing to the change in the original sampling rate, we examined the time-frequency response of some signals by visualizing their spectrograms. As shown in Fig. 22, the behavior of the time-frequency signature is similar to that computed for the Parícutín dataset, within the 1–10 (Hz) frequency range. This allowed us to retrieve the proposed physical features for this dataset, including three amplitude signals for the 1–2 (Hz), 2–4 (Hz), and 4–6 (Hz) bands, as well as the incidence angle.

The selection of consistent frequency bands for analysis, primarily in the 1–10 (Hz) range, is a critical parameter that links the Tancítaro-Parícutín Volcanic Region (TPVR) dataset and the Tacaná generalization dataset. This consistency is geophysically motivated by the similar tectonic environment shared by both seismic-volcanic mechanisms.

Both regions are fundamentally influenced by the subduction of the Cocos Plate beneath the North America Plate. This predominant tectonic interaction results in comparable epicentral distances for regional seismic events originating along the Pacific Ocean. Consequently, the signals propagating to the respective acquisition instrumentation exhibit similar energy characteristics within the selected frequency range. While the Tacaná region introduces the additional complexity of the Caribbean Plate interaction, the 1–10 (Hz) band remains significant for analyzing this overarching tectonic setting.

It is important to highlight that this parameter transferability is specific to similar tectonic regimes. Application of this model to seismically or tectonically distinct regions would necessitate an adaptation of the input frequency bands, and performance metrics (such as prediction accuracy and sensitivity) would likely reflect different degrees of approximation.

Some results from the phase-picking process are shown in Fig. 23. The probability mask for each phase is well determined by the model for both P- and S-wave time arrivals. Both masks exceed the established threshold for minimum probability (>0.5) in the output layer's activation function.

Fig. 19 Signals with their corresponding P-wave arrival time and S-wave arrival time probability are represented by the blue line. The ground truth labels are identified with the green line

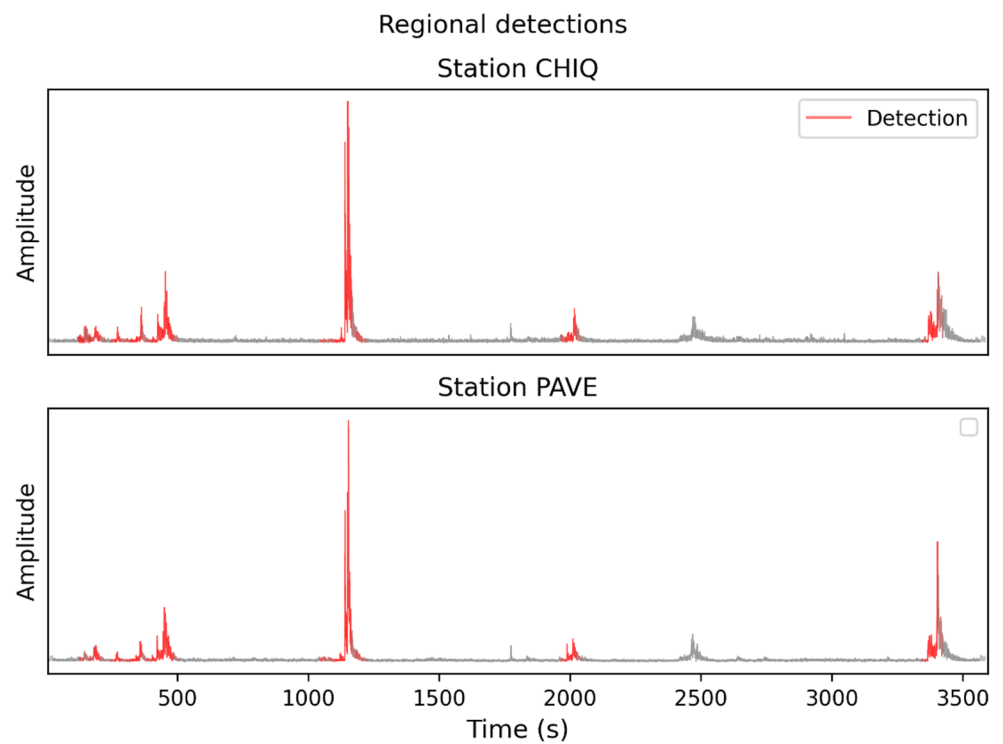
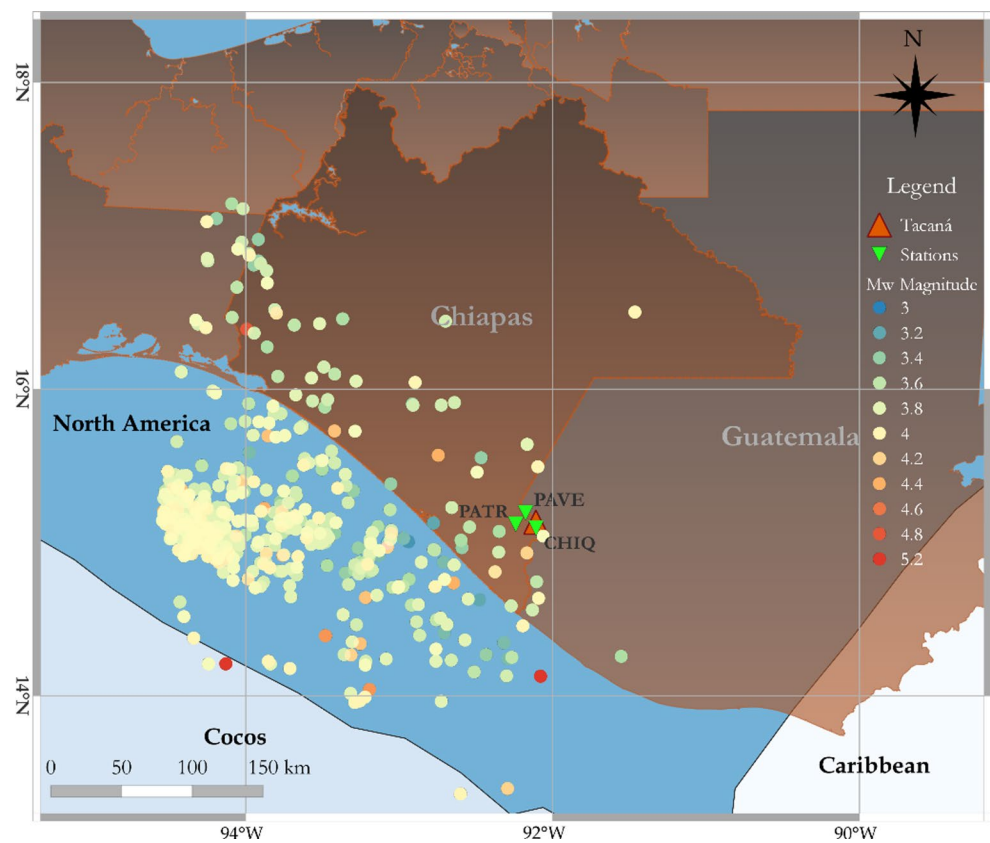


Fig. 20 Regional detections in CHIQ and PAVE stations



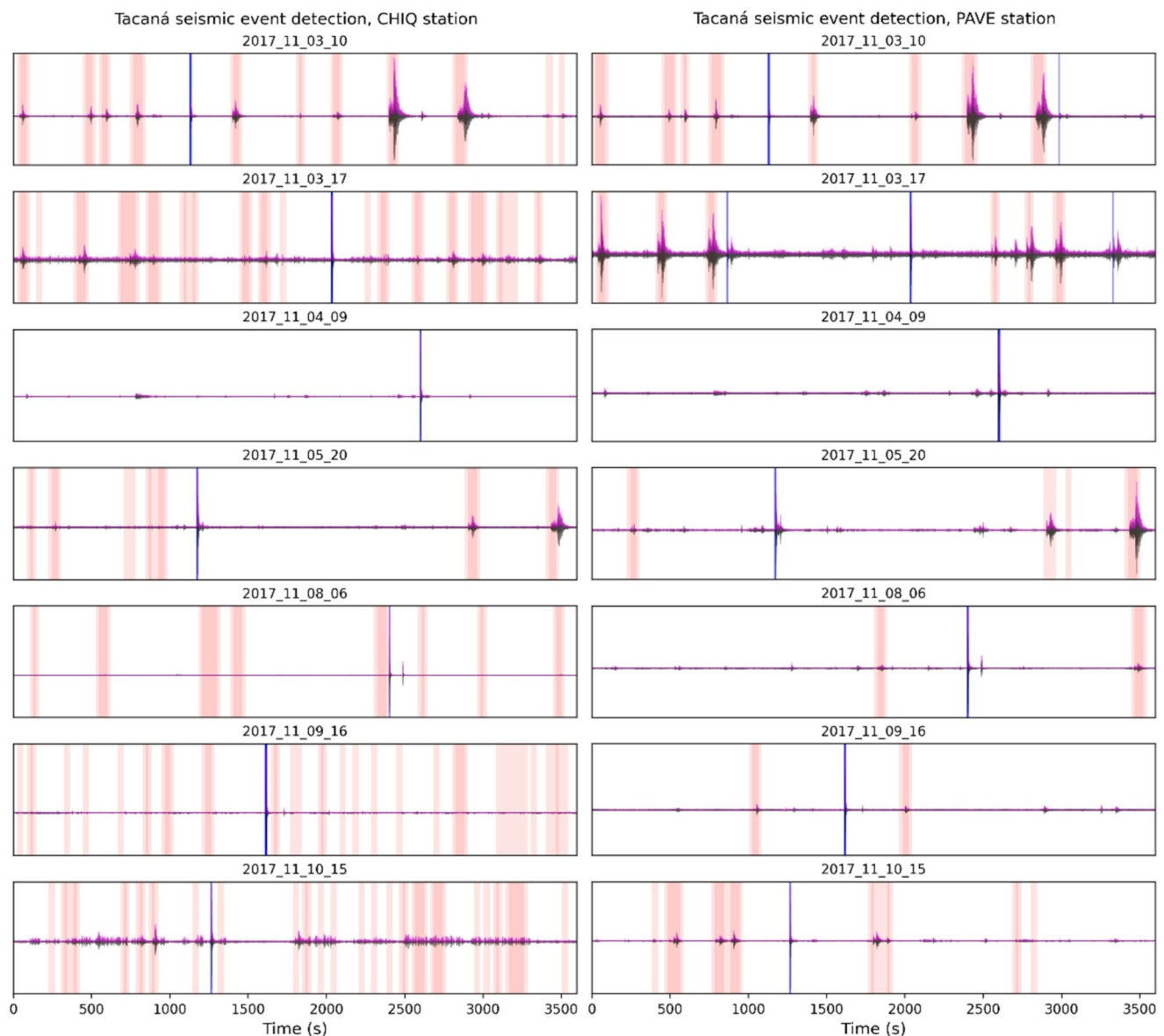


Fig. 21 Seismic activity in the surroundings of Tacaná volcano from 11/01/2017 to 11/30/2017

Table 4 Generalization results for seismic event detection for November 2017. * After the regional detection and removal of the respective samples. ** respect to total number of 10 s windows

Station	CHIQ	PAVE
Total 40 s windows	129,600	129,600
Total 10 s windows*	282,710	432,662
# Local events confirmed	20/22	21/22
# Regional events detected	1630	710
Local false positive (# windows)	241	286
Accuracy rate**	99.914%	99.933%

A total of 589 examples were detected as regional events, with an average TS-TP of 20.139 s. The average probability prediction for P-wave arrival was 0.8364, and the average probability for S-wave arrival was 0.8412. This suggests a stable probability distribution for both wave phases, whereas the average difference in time arrivals indicates that most seismic sources are within a range of 100–300 km from the receivers, assuming typical crustal velocities for P-waves (5.5 km/s) and S-waves (3.2 km/s), corresponding to the average source locations for November 2017 (Fig. 21).

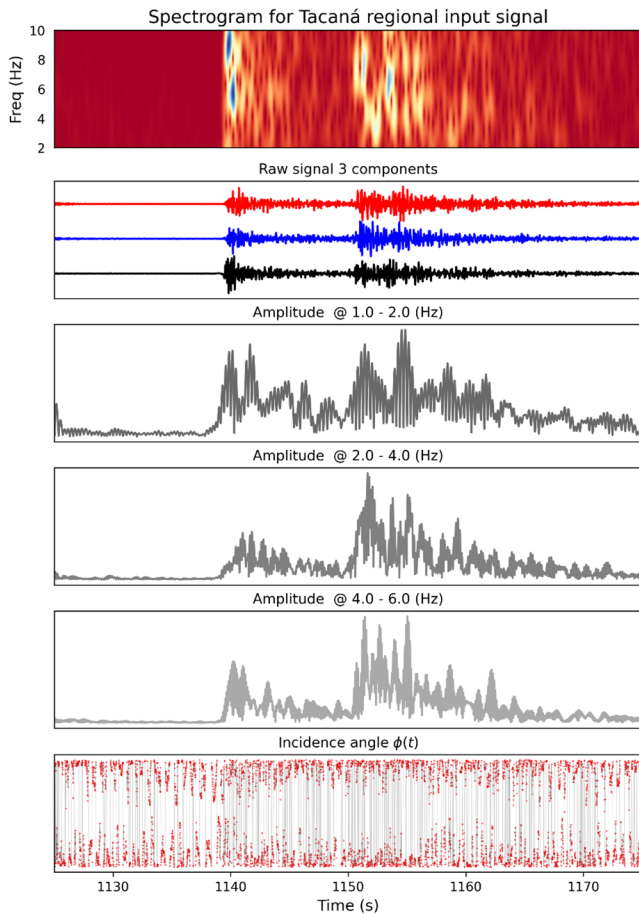
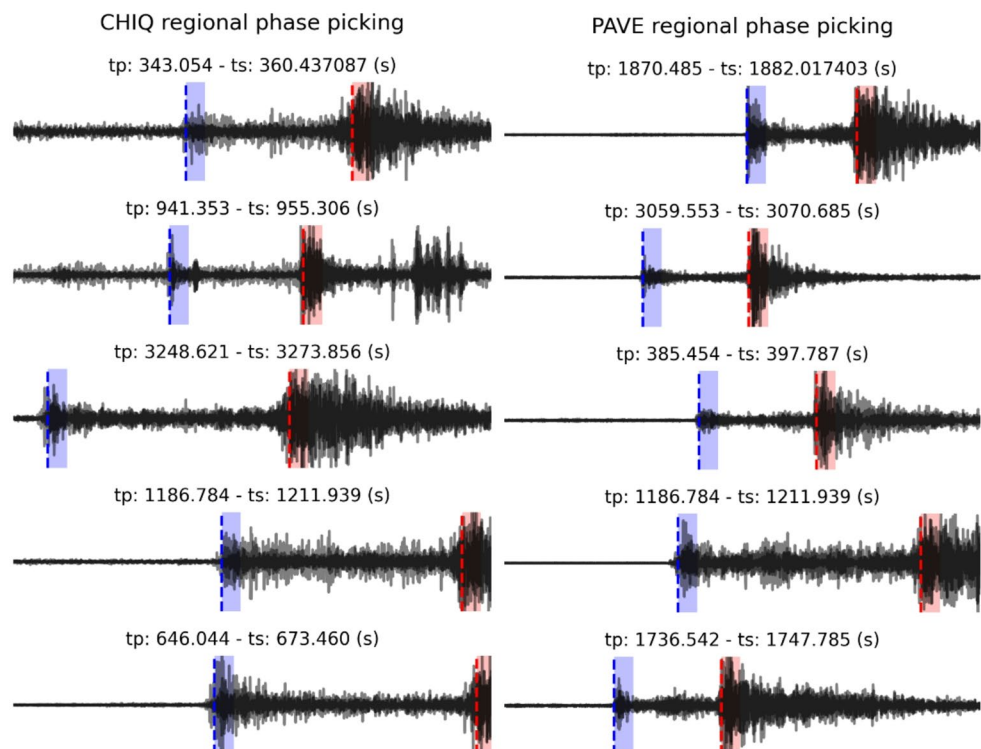


Fig. 22 Physical features of a signal sample of the Tacaná dataset

Fig. 23 Results from regional phase-picking on Tacaná random selected examples. Left column CHIQ station, right column PAVE station. P, S-wave time arrival predictions are blue and red shaded, respectively



Conclusions

Feature Engineering Scheme.

- The proposed feature engineering scheme for developing physically oriented deep learning models demonstrated acceptable performance throughout the training, testing, and generalization stages. This approach is beneficial for implementing statistical learning models when databases are not large enough (i.e., contain at least hundreds of thousands of input samples), as in the case studied in this work. It offers two key advantages: increasing the number of inputs for the training process and refining the feature space by applying appropriate transformations that account for the physical significance of the seismological data.

Seismic Event Detection.

- A simple deep learning model was implemented considering the leverage of a physically driven feature engineering based on the energy envelope of seismic signal components. With this approach the model predicted regional and local events from background noise effectively with reduced computational costs and high accuracy, resulting in a reliable alternative to traditional algorithms.
- Multiblock convolutional neural networks exhibit high performance in discriminating signals from background

noise via a simple feature engineering scheme, such as the use of the energy envelope alone. This approach integrates the 3-component features into a single signature. Overall performance across different sampling rate schemes ranges from sufficient to excellent, as indicated by diverse metrics such as accuracy and the Matthews correlation coefficient for binary classifications. The general evaluation shows that the proposed classifier is effective with completely unseen examples, achieving an accuracy of over 95%. Additionally, the training time is fast, making this approach a promising alternative when limited high-quality data is available.

Seismic Phase Picking.

- Compared with using raw data for training, the proposed encoder-decoder model, which is built on a feature engineering scheme, shows significant improvements. From various suitable evaluation metrics for imbalanced data, it can be concluded that feature engineering enhances the prediction performance for small datasets, with data quality ranging from good to excellent. This approach presents a practical alternative for developing deep learning tools to assist with some of the most essential tasks in seismological data analysis.

Generalization test.

- The seismo-volcanic data is inherently complex, with signal characteristics strongly modulated by both regional tectonic and highly variable local geological processes. For comprehensive interpretation, a clear understanding of this local effect must be integrated into the analysis. Despite these inherent challenges and the data constraints, the generalizability performance of the proposed joint detection and phase-picking models is found to be acceptable. Therefore, the proposed methodology, particularly its reliance on physically driven feature engineering, provides a robust and practical framework for seismic monitoring that can be successfully applied across different, though tectonically similar, volcanic regimes.

Future Work Perspectives.

- There are numerous opportunities for extending this approach. One immediate area of focus is seismic event detection and discrimination for local and regional phenomena. Additionally, integrating the processes of detection and phase-picking to automatically determine the source location could be a valuable advancement.

- Another area for development is creating a feature engineering model for multiple wave phases, beyond just P and S-waveforms. This would help classify derived phases in more complex geological contexts.

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Data availability Must be requested to M. Pertón for Tancitaro - Parícutin seismic data, and to D. Legrand for the Tacaná seismic data.

Code availability Code for testing the trained deep learning models with a small chunk of Tacaná data is available with reasonable request to the corresponding author.

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